Verification Appendix(Coverage Report)

Document Purpose

This appendix contains functional coverage reports generated for all RTL modules of the 5-Stage Pipelined RISC-V Processor.

Verification Environment:
• SystemVerilog Testbench
• Assertions
• Functional Coverage
• Riviera-PRO Simulator

Total Modules Verified : 24
Coverage Goal : 100%
Coverage Achieved : 100%

Table of Contents

1.ALU

```
VSIMSA: Configuration file changed: `/home/runner/library.cfg'
ALIB: Library "work" attached.
work = ./work/work.lib
MESSAGE "Unit top modules: tb_alu."
SUCCESS "Compile success 0 Errors 0 Warnings  Analysis time: 0[s]."
done
# Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.
# HDL, SystemC, and Assertions simulator, debugger, and design environment.
# (c) 1999-2025 Aldec, Inc. All rights reserved.
vsim +access+r
# ELBREAD: Elaboration process.
# ELBREAD: Elaboration time 0.0 [s].
# KERNEL: Main thread initiated.
# KERNEL: Kernel process initialization phase.
# ELAB2: Elaboration final pass...
# KERNEL: PLI/VHPI kernel's engine initialization done.
# PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'
# ELAB2: Create instances ...
# KERNEL: Time resolution set to 1ps.
# ELAB2: Create instances complete.
# SLP: Started
# SLP: Elaboration phase ...
# SLP: Elaboration phase ... done : 0.0 [s]
# SLP: Generation phase ...
# SLP: Generation phase ... done : 0.1 [s]
# SLP: Finished : 0.1 [s]
# SLP: 0 primitives and 1 (33.33%) other processes in SLP
# SLP: 5 (12.82%) signals in SLP and 5 (12.82%) interface signals
# ELAB2: Elaboration final pass complete - time: 0.2 [s].
# KERNEL: SLP loading done - time: 0.0 [s].
```

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5328 kB (elbread=455 elab2=4862 kernel=11 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : ADD

KERNEL: PASS : ADD

KERNEL: ASSERT PASS : ADD

KERNEL:

KERNEL: TC2 : SUB

KERNEL: PASS : SUB

KERNEL: ASSERT PASS : SUB

KERNEL:

KERNEL: TC3 : AND

KERNEL: PASS : AND

KERNEL: ASSERT PASS : AND

KERNEL:

KERNEL: TC4 : OR

KERNEL: PASS : OR

KERNEL: ASSERT PASS : OR

KERNEL:

KERNEL: TC5 : XOR

KERNEL: PASS : XOR

KERNEL: ASSERT PASS : XOR

KERNEL:

KERNEL: TC6 : SLL

KERNEL: PASS : SLL

KERNEL: ASSERT PASS : SLL

KERNEL:

KERNEL: TC7 : SRL

KERNEL: PASS : SRL

KERNEL: ASSERT PASS : SRL

KERNEL:

KERNEL: TC8 : SRA

KERNEL: PASS : SRA

KERNEL: ASSERT PASS : SRA

KERNEL:

KERNEL: TC9 : SLT

KERNEL: PASS : SLT

KERNEL: ASSERT PASS : SLT

KERNEL:

KERNEL: TC10 : SLTU

KERNEL: PASS : SLTU

KERNEL: ASSERT PASS : SLTU

KERNEL:

KERNEL: TC11 : LOAD

KERNEL: PASS : LOAD

KERNEL: ASSERT PASS : LOAD

KERNEL:

KERNEL: TC12 : BEQ

KERNEL: PASS : BEQ

KERNEL: ASSERT PASS : BEQ

KERNEL:

KERNEL: TC13 : BNE

KERNEL: PASS : BNE

KERNEL: ASSERT PASS : BNE

KERNEL:

KERNEL: TC14 : BLT

KERNEL: PASS : BLT

KERNEL: ASSERT PASS : BLT

KERNEL:

```
# KERNEL: TC15 : BGE
# KERNEL: PASS : BGE
# KERNEL: ASSERT PASS : BGE
# KERNEL:
# KERNEL: TC16 : LUI
# KERNEL: PASS : LUI
# KERNEL: ASSERT PASS : LUI
# KERNEL:
# KERNEL: TC17 : JAL
# KERNEL: PASS : JAL
# KERNEL: ASSERT PASS : JAL
# KERNEL:
# KERNEL: TC18 : DEFAULT
# KERNEL: PASS : DEFAULT
# KERNEL: ASSERT PASS : DEFAULT
# KERNEL:
# KERNEL: TC19 : ADDI
# KERNEL: PASS : ADDI
# KERNEL: ASSERT PASS : ADDI
# KERNEL:
# KERNEL: TC20 : BLTU
# KERNEL: PASS : BLTU
# KERNEL: ASSERT PASS : BLTU
# KERNEL:
# KERNEL: TC21 : BGEU
# KERNEL: PASS : BGEU
# KERNEL: ASSERT PASS : BGEU
# KERNEL:
# KERNEL: ALU VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (553): $finish called.
```

```
# KERNEL: Time: 115 ns,  Iteration: 0,  Instance: /tb_alu,  Process: @INITIAL#85_0@.
# KERNEL: stopped at time: 115 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | 67440b0adff8             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-18 07:24         |
# | Report arguments | -verbose                 |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 07:24         |
# | Number of tests  | 1                        |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |              Value              |
# ==============================================
```

```
# | Test     | fcover.acdb:fcover             |
# | Status   | Ok                             |
# | Args     | asim +access+r                 |
# | Simtime  | 115000 ps                      |
# | Cputime  | 0.057 s                        |
# | Seed     | 1                              |
# | Date     | 2026-06-18 07:24               |
# | User     | runner                         |
# | Host     | 67440b0adff8                   |
# | Host os  | Linux64                        |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
```

```
# INSTANCE - /tb_alu : work.tb_alu
#
#
#    SUMMARY
#
=====================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
=====================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
=====================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
=====================================================================
==
#    |              Covergroup              |   Hits   |  Goal /  |  Status  |
#    |                                      |          | At Least |          |
#
=====================================================================
==
#    | TYPE /tb_alu/alu_cg                  | 100.000% | 100.000% | Covered  |
#
=====================================================================
==
#    | INSTANCE ALU_COVERAGE                | 100.000% | 100.000% | Covered  |
#    |--------------------------------------|----------|----------|----------|
#    | COVERPOINT ALU_COVERAGE::cp_operation | 100.000% | 100.000% | Covered  |
```

```
#     |--------------------------------------|----------|----------|----------|
#     | bin add_bin                          |        1 |        1 | Covered  |
#     | bin sub_bin                          |        1 |        1 | Covered  |
#     | bin sll_bin                          |        1 |        1 | Covered  |
#     | bin srl_bin                          |        1 |        1 | Covered  |
#     | bin sra_bin                          |        1 |        1 | Covered  |
#     | bin slt_bin                          |        1 |        1 | Covered  |
#     | bin sltu_bin                         |        1 |        1 | Covered  |
#     | bin xor_bin                          |        1 |        1 | Covered  |
#     | bin or_bin                           |        1 |        1 | Covered  |
#     | bin and_bin                          |        1 |        1 | Covered  |
#     | bin addi_bin                         |        1 |        1 | Covered  |
#     | bin loadstore_bin                    |        1 |        1 | Covered  |
#     | bin beq_bin                          |        1 |        1 | Covered  |
#     | bin bne_bin                          |        1 |        1 | Covered  |
#     | bin blt_bin                          |        1 |        1 | Covered  |
#     | bin bge_bin                          |        1 |        1 | Covered  |
#     | bin bltu_bin                         |        1 |        1 | Covered  |
#     | bin bgeu_bin                         |        1 |        1 | Covered  |
#     | bin lui_bin                          |        1 |        1 | Covered  |
#     | bin jal_bin                          |        1 |        1 | Covered  |
#     | default bin default_bin              |        1 |    -     | Occurred |
#
==============================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
# +++++++++++       DESIGN UNITS      +++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
```

```
# ==================================================
# |    Coverage Type    | Weight | Hits/Total |
# ==================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_alu
#    SUMMARY
#    ==================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ==================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ==================================================
#    WEIGHTED AVERAGE: 100.000%
#    COVERGROUP COVERAGE
#
==============================================================================
==
#    |            Covergroup             |   Hits   |  Goal /  |  Status  |
#    |                                   |          | At Least |          |
#
==============================================================================
==
#    | TYPE /tb_alu/alu_cg               | 100.000% | 100.000% | Covered
==============================================================================
==
```

```
#     | INSTANCE ALU_COVERAGE                | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | COVERPOINT ALU_COVERAGE::cp_operation | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | bin add_bin                          |        1 |        1 | Covered  |
#     | bin sub_bin                          |        1 |        1 | Covered  |
#     | bin sll_bin                          |        1 |        1 | Covered  |
#     | bin srl_bin                          |        1 |        1 | Covered  |
#     | bin sra_bin                          |        1 |        1 | Covered  |
#     | bin slt_bin                          |        1 |        1 | Covered  |
#     | bin sltu_bin                         |        1 |        1 | Covered  |
#     | bin xor_bin                          |        1 |        1 | Covered  |
#     | bin or_bin                           |        1 |        1 | Covered  |
#     | bin and_bin                          |        1 |        1 | Covered  |
#     | bin addi_bin                         |        1 |        1 | Covered  |
#     | bin loadstore_bin                    |        1 |        1 | Covered  |
#     | bin beq_bin                          |        1 |        1 | Covered  |
#     | bin bne_bin                          |        1 |        1 | Covered  |
#     | bin blt_bin                          |        1 |        1 | Covered  |
#     | bin bge_bin                          |        1 |        1 | Covered  |
#     | bin bltu_bin                         |        1 |        1 | Covered  |
#     | bin bgeu_bin                         |        1 |        1 | Covered  |
#     | bin lui_bin                          |        1 |        1 | Covered  |
#     | bin jal_bin                          |        1 |        1 | Covered  |
#     | default bin default_bin              |        1 |    -     | Occurred |
#
==========================================================================
==
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

2. ALU_Control

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_alu_control."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 4 (10.81%) signals in SLP and 4 (10.81%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5318 kB (elbread=459 elab2=4848 kernel=10 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : LOAD STORE

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : LOAD STORE

KERNEL:

KERNEL: TC2 : BEQ

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : BEQ

KERNEL:

KERNEL: TC3 : BNE

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : BNE

KERNEL:

KERNEL: TC4 : BLT

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : BLT

KERNEL:

KERNEL: TC5 : BGE

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : BGE

KERNEL:

KERNEL: TC6 : ADD

KERNEL: PASS : TC6

KERNEL: ASSERT PASS : ADD

KERNEL:

KERNEL: TC7 : SUB

KERNEL: PASS : TC7

KERNEL: ASSERT PASS : SUB

KERNEL:

KERNEL: TC8 : SLL

KERNEL: PASS : TC8

KERNEL: ASSERT PASS : SLL

KERNEL:

KERNEL: TC9 : XOR

KERNEL: PASS : TC9

KERNEL: ASSERT PASS : XOR

KERNEL:

KERNEL: TC10 : SRA

KERNEL: PASS : TC10

KERNEL: ASSERT PASS : SRA

KERNEL:

KERNEL: TC11 : ADDI

KERNEL: PASS : TC11

KERNEL: ASSERT PASS : ADDI

KERNEL:

KERNEL: TC12 : SLLI

KERNEL: PASS : TC12

KERNEL: ASSERT PASS : SLLI

KERNEL:

KERNEL: TC13 : SRLI

KERNEL: PASS : TC13

KERNEL: ASSERT PASS : SRLI

KERNEL:

KERNEL: TC14 : JAL

KERNEL: PASS : TC14

KERNEL: ASSERT PASS : JAL

KERNEL:

```
# KERNEL: TC15 : JALR
# KERNEL: PASS : TC15
# KERNEL: ASSERT PASS : JALR
# KERNEL:
# KERNEL: TC16 : LUI
# KERNEL: PASS : TC16
# KERNEL: ASSERT PASS : LUI
# KERNEL:
# KERNEL: TC17 : AUIPC
# KERNEL: PASS : TC17
# KERNEL: ASSERT PASS : AUIPC
# KERNEL:
# KERNEL: ALU_CONTROL VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (416): $finish called.
# KERNEL: Time: 95 ns,  Iteration: 0,  Instance: /tb_alu_control,  Process: @INITIAL#55_0@.
# KERNEL: stopped at time: 95 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++      REPORT INFO      ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property     |          Value            |
# ===============================================
```

```
# | User            | runner                |
# | Host            | 3e88754d5dab          |
# | Tool            | Riviera-PRO 2025.04   |
# | Report file     | /home/runner/cov.txt  |
# | Report date     | 2026-06-18 16:05      |
# | Report arguments | -verbose             |
# | Input file      | /home/runner/fcover.acdb |
# | Input file date | 2026-06-18 16:05      |
# | Number of tests | 1                     |
# ==================================================
#
#
# TEST DETAILS
# ==================================================
# | Property |            Value               |
# ==================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 95000 ps                        |
# | Cputime  | 0.037 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-18 16:05                |
# | User     | runner                          |
# | Host     | 3e88754d5dab                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++    DESIGN HIERARCHY    ++++++++++
```

```
# ++++++++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =====================================================
# |    Coverage Type    | Weight | Hits/Total |
# =====================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =====================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_alu_control : work.tb_alu_control
#
#
#    SUMMARY
#
=================================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
=================================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
=================================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
```

```
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===========================================================================
=====
#    |              Covergroup              |   Hits   |  Goal /  | Status  |
#    |                                      |          | At Least |         |
#
===========================================================================
=====
#    | TYPE /tb_alu_control/alu_control_cg      | 100.000% | 100.000% | Covered |
#
===========================================================================
=====
#    | INSTANCE ALU_CONTROL_COVERAGE            | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT ALU_CONTROL_COVERAGE::cp_aluop | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin load_store                           |        1 |        1 | Covered |
#    | bin branch                               |        1 |        1 | Covered |
#    | bin r_type                               |        1 |        1 | Covered |
#    | bin i_type                               |        1 |        1 | Covered |
#    | bin jal                                  |        1 |        1 | Covered |
#    | bin jalr                                 |        1 |        1 | Covered |
#    | bin lui                                  |        1 |        1 | Covered |
#    | bin auipc                                |        1 |        1 | Covered |
#
===========================================================================
=====
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS       ++++++++++
```

```
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_alu_control
#
#
#    SUMMARY
#    ================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
```

```
#
=========================================================================
=====
#    |                 Covergroup                 |   Hits   |  Goal /  | Status  |
#    |                                            |          | At Least |         |
#
=========================================================================
=====
#    | TYPE /tb_alu_control/alu_control_cg       | 100.000% | 100.000% | Covered |
#
=========================================================================
=====
#    | INSTANCE ALU_CONTROL_COVERAGE             | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT ALU_CONTROL_COVERAGE::cp_aluop | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin load_store                           |        1 |        1 | Covered |
#    | bin branch                               |        1 |        1 | Covered |
#    | bin r_type                               |        1 |        1 | Covered |
#    | bin i_type                               |        1 |        1 | Covered |
#    | bin jal                                  |        1 |        1 | Covered |
#    | bin jalr                                 |        1 |        1 | Covered |
#    | bin lui                                  |        1 |        1 | Covered |
#    | bin auipc                                |        1 |        1 | Covered |
#
=========================================================================
=====
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

3.ADDER

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_adder."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 2 (6.06%) signals in SLP and 2 (6.06%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5294 kB (elbread=455 elab2=4830 kernel=8 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : ZERO

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : ZERO

KERNEL:

KERNEL: TC2 : SMALL VALUE

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : SMALL

KERNEL:

KERNEL: TC3 : MEDIUM VALUE

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : MEDIUM

KERNEL:

KERNEL: TC4 : LARGE VALUE

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : LARGE

KERNEL:

KERNEL: TC5 : MAX VALUE

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : OVERFLOW

KERNEL:

KERNEL: ADDER VERIFICATION COMPLETE

KERNEL:

KERNEL: Coverage = 100.00%

RUNTIME: Info: RUNTIME_0068 testbench.sv (163): $finish called.

```
# KERNEL: Time: 35 ns,  Iteration: 0,  Instance: /tb_adder,  Process: @INITIAL#50_0@.
# KERNEL: stopped at time: 35 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++      REPORT INFO      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value          |
# ===============================================
# | User             | runner                  |
# | Host             | e81aa665321a            |
# | Tool             | Riviera-PRO 2025.04     |
# | Report file      | /home/runner/cov.txt    |
# | Report date      | 2026-06-18 15:15        |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 15:15        |
# | Number of tests  | 1                       |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |             Value             |
# ==============================================
```

```
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 35000 ps                        |
# | Cputime  | 0.152 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-18 15:15                |
# | User     | runner                          |
# | Host     | e81aa665321a                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ===============================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
```

```
# INSTANCE - /tb_adder : work.tb_adder
#
#
#    SUMMARY
#
==========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
==========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
==========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#    ========================================================================
#    |            Covergroup            |   Hits   |  Goal /  | Status  |
#    |                                  |          | At Least |         |
#    ========================================================================
#    | TYPE /tb_adder/adder_cg          | 100.000% | 100.000% | Covered |
#    ========================================================================
#    | INSTANCE ADDER_COVERAGE          | 100.000% | 100.000% | Covered |
#    |----------------------------------|----------|----------|---------|
#    | COVERPOINT ADDER_COVERAGE::cp_pc | 100.000% | 100.000% | Covered |
#    |---------------------------------|----------|----------|---------|
#    | bin zero_val                     |        1 |        1 | Covered |
#    | bin small_val                    |        1 |        1 | Covered |
```

```
#     | bin medium_val                    |         1 |         1 | Covered |
#     | bin large_val                     |         1 |         1 | Covered |
#     | bin max_val                       |         1 |         1 | Covered |
#     ==========================================================================
#
#
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_adder
#
#
#     SUMMARY
#     =============================================
#     |    Coverage Type    | Weight | Hits/Total |
#     =============================================
#     | Covergroup Coverage |      1 |   100.000% |
```

```
#    |--------------------|--------|------------|
#    | Types              |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#    ==========================================================================
#    |          Covergroup          |   Hits   |  Goal /  | Status  |
#    |                              |          | At Least |         |
#    ==========================================================================
#    | TYPE /tb_adder/adder_cg      | 100.000% | 100.000% | Covered |
#    ==========================================================================
#    | INSTANCE ADDER_COVERAGE      | 100.000% | 100.000% | Covered |
#    |--------------------------------|----------|----------|---------|
#    | COVERPOINT ADDER_COVERAGE::cp_pc | 100.000% | 100.000% | Covered |
#    |--------------------------------|----------|----------|---------|
#    | bin zero_val                 |        1 |        1 | Covered |
#    | bin small_val                |        1 |        1 | Covered |
#    | bin medium_val               |        1 |        1 | Covered |
#    | bin large_val                |        1 |        1 | Covered |
#    | bin max_val                  |        1 |        1 | Covered |
#    ==========================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished
```

4. Branch Decision

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_branch_decision."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 3 (60.00%) other processes in SLP

SLP: 5 (5.49%) signals in SLP and 5 (5.49%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6357 kB (elbread=463 elab2=5883 kernel=10 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1

KERNEL: ASSERT PASS : IDLE

KERNEL: PASS : TC1

KERNEL:

KERNEL: TC2

KERNEL: ASSERT PASS : IDLE

KERNEL: PASS : TC2

KERNEL:

KERNEL: TC3

KERNEL: ASSERT PASS : BRANCH NOT TAKEN

KERNEL: PASS : TC3

KERNEL:

KERNEL: TC4

KERNEL: ASSERT PASS : BRANCH TAKEN

KERNEL: PASS : TC4

KERNEL:

KERNEL: TC5

KERNEL: ASSERT PASS : JUMP

KERNEL: PASS : TC5

KERNEL:

KERNEL: TC6

KERNEL: ASSERT PASS : JUMP

KERNEL: PASS : TC6

KERNEL:

```
# KERNEL: TC7
# KERNEL: ASSERT PASS : JUMP
# KERNEL: PASS : TC7
# KERNEL:
# KERNEL: TC8
# KERNEL: ASSERT PASS : JUMP
# KERNEL: PASS : TC8
# KERNEL:
# KERNEL: BRANCH_DECISION VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (280): $finish called.
# KERNEL: Time: 50 ns,  Iteration: 0,  Instance: /tb_branch_decision,  Process: @INITIAL#115_1@.
# KERNEL: stopped at time: 50 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | 254dc240d4ec             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
```

```
# | Report date      | 2026-06-18 12:47        |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 12:47        |
# | Number of tests  | 1                       |
# ==================================================
#
#
# TEST DETAILS
# =================================================
# | Property |             Value              |
# =================================================
# | Test     | fcover.acdb:fcover             |
# | Status   | Ok                             |
# | Args     | asim +access+r                 |
# | Simtime  | 50000 ps                       |
# | Cputime  | 0.028 s                        |
# | Seed     | 1                              |
# | Date     | 2026-06-18 12:47               |
# | User     | runner                         |
# | Host     | 254dc240d4ec                   |
# | Host os  | Linux64                        |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# =================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
```

```
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
# INSTANCE - /tb_branch_decision : work.tb_branch_decision
#     SUMMARY
#
=============================================================================
==
#     |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
=============================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
=============================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
=============================================================================
================
#     |                    Covergroup                    |   Hits   |  Goal /  | Status  |
#     |                                                  |          | At Least |         |
```

#
===

| TYPE /tb_branch_decision/branch_decision_cg | 100.000% | 100.000% | Covered |

#
===

| INSTANCE BRANCH_DECISION_COVERAGE | 100.000% | 100.000% | Covered |

|---|----------|----------|---------|

| COVERPOINT BRANCH_DECISION_COVERAGE::cp_branch_condn | 100.000% | 100.000% | Covered |

|---|----------|----------|---------|

| bin zero | 4 | 1 | Covered |

| bin one | 4 | 1 | Covered |

|---|----------|----------|---------|

| COVERPOINT BRANCH_DECISION_COVERAGE::cp_branch | 100.000% | 100.000% | Covered |

|---|----------|----------|---------|

| bin zero | 4 | 1 | Covered |

| bin one | 4 | 1 | Covered |

|---|----------|----------|---------|

| COVERPOINT BRANCH_DECISION_COVERAGE::cp_jump | 100.000% | 100.000% | Covered |

|---|----------|----------|---------|

| bin zero | 4 | 1 | Covered |

| bin one | 4 | 1 | Covered |

|---|----------|----------|---------|

| COVERPOINT BRANCH_DECISION_COVERAGE::cp_pcsrc | 100.000% | 100.000% | Covered |

|---|----------|----------|---------|

| bin zero | 3 | 1 | Covered |

| bin one | 5 | 1 | Covered |

|---|----------|----------|---------|

| CROSS BRANCH_DECISION_COVERAGE::cross_all | 100.000% | 100.000% | Covered |

```
#     |----------------------------------------------------|----------|----------|---------|
#     | bin <zero,zero,zero>                               |        1 |        1 | Covered |
#     | bin <zero,zero,one>                                |        1 |        1 | Covered |
#     | bin <zero,one,zero>                                |        1 |        1 | Covered |
#     | bin <zero,one,one>                                 |        1 |        1 | Covered |
#     | bin <one,zero,zero>                                |        1 |        1 | Covered |
#     | bin <one,zero,one>                                 |        1 |        1 | Covered |
#     | bin <one,one,zero>                                 |        1 |        1 | Covered |
#     | bin <one,one,one>                                  |        1 |        1 | Covered |
#
================================================================================================
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++      DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
# CUMULATIVE SUMMARY
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_branch_decision
#
#
#     SUMMARY
#     ================================================
```

```
#     |    Coverage Type     | Weight | Hits/Total |
#     ===============================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |---------------------|--------|------------|
#     | Types               |        |      1 / 1 |
#     ===============================================
#     WEIGHTED AVERAGE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
=============================================================================
================
#     |                    Covergroup                    |   Hits   |  Goal /  | Status  |
#     |                                                  |          | At Least |         |
#
=============================================================================
================
#     | TYPE /tb_branch_decision/branch_decision_cg         | 100.000% | 100.000% | Covered |
#
=============================================================================
================
#     | INSTANCE BRANCH_DECISION_COVERAGE                  | 100.000% | 100.000% | Covered
|
#     |--------------------------------------------------|----------|----------|---------|
#     | COVERPOINT BRANCH_DECISION_COVERAGE::cp_branch_condn | 100.000% |
100.000% | Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | bin zero                                        |        4 |        1 | Covered |
#     | bin one                                         |        4 |        1 | Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | COVERPOINT BRANCH_DECISION_COVERAGE::cp_branch       | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | bin zero                                        |        4 |        1 | Covered |
```

```
#     | bin one                                          |         4 |         1 | Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | COVERPOINT BRANCH_DECISION_COVERAGE::cp_jump         | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | bin zero                                         |         4 |         1 | Covered |
#     | bin one                                          |         4 |         1 | Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | COVERPOINT BRANCH_DECISION_COVERAGE::cp_pcsrc        | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | bin zero                                         |         3 |         1 | Covered |
#     | bin one                                          |         5 |         1 | Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | CROSS BRANCH_DECISION_COVERAGE::cross_all            | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------------|----------|----------|---------|
#     | bin <zero,zero,zero>                             |         1 |         1 | Covered |
#     | bin <zero,zero,one>                              |         1 |         1 | Covered |
#     | bin <zero,one,zero>                              |         1 |         1 | Covered |
#     | bin <zero,one,one>                               |         1 |         1 | Covered |
#     | bin <one,zero,zero>                              |         1 |         1 | Covered |
#     | bin <one,zero,one>                               |         1 |         1 | Covered |
#     | bin <one,one,zero>                               |         1 |         1 | Covered |
#     | bin <one,one,one>                                |         1 |         1 | Covered |
#
=====================================================================================
================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

5. Control

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_control."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.0 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 10 (20.41%) signals in SLP and 10 (20.41%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5311 kB (elbread=459 elab2=4843 kernel=9 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : R_TYPE

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : R_TYPE

KERNEL:

KERNEL: TC2 : I_TYPE

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : I_TYPE

KERNEL:

KERNEL: TC3 : LOAD

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : LOAD

KERNEL:

KERNEL: TC4 : STORE

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : STORE

KERNEL:

KERNEL: TC5 : BRANCH

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : BRANCH

KERNEL:

KERNEL: TC6 : JAL

KERNEL: PASS : TC6

```
# KERNEL: ASSERT PASS : JAL
# KERNEL:
# KERNEL: TC7 : JALR
# KERNEL: PASS : TC7
# KERNEL: ASSERT PASS : JALR
# KERNEL:
# KERNEL: TC8 : LUI
# KERNEL: PASS : TC8
# KERNEL: ASSERT PASS : LUI
# KERNEL:
# KERNEL: TC9 : AUIPC
# KERNEL: PASS : TC9
# KERNEL: ASSERT PASS : AUIPC
# KERNEL:
# KERNEL: TC10 : INVALID
# KERNEL: PASS : TC10
# KERNEL: ASSERT PASS : DEFAULT
# KERNEL:
# KERNEL: CONTROL VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (266): $finish called.
# KERNEL: Time: 60 ns,  Iteration: 0,  Instance: /tb_control,  Process: @INITIAL#64_0@.
# KERNEL: stopped at time: 60 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
```

```
#
#
# SUMMARY
# ==================================================
# |     Property     |          Value          |
# ==================================================
# | User             | runner                  |
# | Host             | 74205c6da5ab            |
# | Tool             | Riviera-PRO 2025.04     |
# | Report file      | /home/runner/cov.txt    |
# | Report date      | 2026-06-18 16:14        |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 16:14        |
# | Number of tests  | 1                       |
# ==================================================
#
#
# TEST DETAILS
# =================================================
# | Property |             Value             |
# =================================================
# | Test     | fcover.acdb:fcover            |
# | Status   | Ok                            |
# | Args     | asim +access+r                |
# | Simtime  | 60000 ps                      |
# | Cputime  | 0.025 s                       |
# | Seed     | 1                             |
# | Date     | 2026-06-18 16:14              |
# | User     | runner                        |
# | Host     | 74205c6da5ab                  |
# | Host os  | Linux64                       |
```

```
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ===============================================
# |    Coverage Type    | Weight | Hits/Total |
# ===============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ===============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
# INSTANCE - /tb_control : work.tb_control
#    SUMMARY
===========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
===========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
===========================================================================
==
```

```
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#     COVERGROUP COVERAGE
#
=========================================================================
===
#     |              Covergroup              |   Hits   |  Goal /  |  Status  |
#     |                                      |          | At Least |          |
#
=========================================================================
===
#     | TYPE /tb_control/control_cg          | 100.000% | 100.000% | Covered  |
#
=========================================================================
===
#     | INSTANCE CONTROL_COVERAGE            | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | COVERPOINT CONTROL_COVERAGE::cp_opcode | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | bin r_type                           |        1 |        1 | Covered  |
#     | bin i_type                           |        1 |        1 | Covered  |
#     | bin load                             |        1 |        1 | Covered  |
#     | bin store                            |        1 |        1 | Covered  |
#     | bin branch                           |        1 |        1 | Covered  |
#     | bin jal                              |        1 |        1 | Covered  |
#     | bin jalr                             |        1 |        1 | Covered  |
#     | bin lui                              |        1 |        1 | Covered  |
#     | bin auipc                            |        1 |        1 | Covered  |
#     | default bin invalid                  |        1 |    -     | Occurred |
#
=========================================================================
===
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
```

```
# CUMULATIVE SUMMARY
# ============================================
# |    Coverage Type    | Weight | Hits/Total |
# ============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_control
#
#
#    SUMMARY
#    ============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ============================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===============================================================================
#    |             Covergroup             |   Hits   |  Goal /  |  Status  |
#    |                                    |          | At Least |          |
```

```
#
===================================================================
===
#    | TYPE /tb_control/control_cg           | 100.000% | 100.000% | Covered  |
#
===================================================================
===
#    | INSTANCE CONTROL_COVERAGE             | 100.000% | 100.000% | Covered  |
#    |--------------------------------------|----------|----------|----------|
#    | COVERPOINT CONTROL_COVERAGE::cp_opcode | 100.000% | 100.000% | Covered  |
#    |--------------------------------------|----------|----------|----------|
#    | bin r_type                           |        1 |        1 | Covered  |
#    | bin i_type                           |        1 |        1 | Covered  |
#    | bin load                             |        1 |        1 | Covered  |
#    | bin store                            |        1 |        1 | Covered  |
#    | bin branch                           |        1 |        1 | Covered  |
#    | bin jal                              |        1 |        1 | Covered  |
#    | bin jalr                             |        1 |        1 | Covered  |
#    | bin lui                              |        1 |        1 | Covered  |
#    | bin auipc                            |        1 |        1 | Covered  |
#    | default bin invalid                  |        1 |    -     | Occurred |
===================================================================
==
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

6. Control Mux

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_control_mux."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 19 (28.36%) signals in SLP and 19 (28.36%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5303 kB (elbread=459 elab2=4835 kernel=8 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : NORMAL FLOW

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : NORMAL FLOW

KERNEL:

KERNEL: TC2 : STALL NOP

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : STALL NOP

KERNEL:

KERNEL: TC3 : DYNAMIC SWITCHING

KERNEL: ASSERT PASS : NORMAL

KERNEL: ASSERT PASS : STALL

KERNEL: PASS : TC3

KERNEL:

KERNEL: CONTROL_MUX VERIFICATION COMPLETE

KERNEL:

KERNEL: Coverage = 100.00%

RUNTIME: Info: RUNTIME_0068 testbench.sv (181): $finish called.

KERNEL: Time: 30 ns, Iteration: 0, Instance: /tb_control_mux, Process: @INITIAL#78_0@.

KERNEL: stopped at time: 30 ns

VSIM: Simulation has finished. There are no more test vectors to simulate.

acdb save

acdb report -db fcover.acdb -txt -o cov.txt -verbose

ACDB: Coverage report "cov.txt" was generated successfully.

```
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value          |
# ===============================================
# | User             | runner                  |
# | Host             | 8076315ac2c3            |
# | Tool             | Riviera-PRO 2025.04     |
# | Report file      | /home/runner/cov.txt    |
# | Report date      | 2026-06-18 16:20        |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 16:20        |
# | Number of tests  | 1                       |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |             Value             |
# ==============================================
# | Test     | fcover.acdb:fcover            |
# | Status   | Ok                            |
# | Args     | asim +access+r                |
# | Simtime  | 30000 ps                      |
# | Cputime  | 0.032 s                       |
# | Seed     | 1                             |
```

```
# | Date     | 2026-06-18 16:20              |
# | User     | runner                        |
# | Host     | 8076315ac2c3                  |
# | Host os  | Linux64                       |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==============================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ============================================
# |    Coverage Type    | Weight | Hits/Total |
# ============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_control_mux : work.tb_control_mux
#
#
#     SUMMARY
#
======================================================================
==
```

```
#     |    Coverage Type     | Weight | Local Hits/Total | Recursive Hits/Total |
#
======================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
======================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
======================================================================
===
#     |               Covergroup               |   Hits   |  Goal /  | Status  |
#     |                                        |          | At Least |         |
#
======================================================================
===
#     | TYPE /tb_control_mux/control_mux_cg     | 100.000% | 100.000% | Covered |
#
======================================================================
===
#     | INSTANCE CONTROL_MUX_COVERAGE          | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT CONTROL_MUX_COVERAGE::cp_sel | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin normal_flow                        |        2 |        1 | Covered |
#     | bin stall_nop                          |        2 |        1 | Covered |
#
======================================================================
===
# ++++++++++++++++++++++++++++++++++++++++++++++++
```

```
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
# MODULE - work.tb_control_mux
#    SUMMARY
#    =============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    =============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    =============================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
=======================================================================
===
#    |              Covergroup              |   Hits   |  Goal /  | Status  |
#    |                                      |          | At Least |         |
#
=======================================================================
===
```

```
#     | TYPE /tb_control_mux/control_mux_cg     | 100.000% | 100.000% | Covered |
#
===================================================================================
#     | INSTANCE CONTROL_MUX_COVERAGE           | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT CONTROL_MUX_COVERAGE::cp_sel | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin normal_flow                        |        2 |        1 | Covered |
#     | bin stall_nop                          |        2 |        1 | Covered |
#
===================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

7. Data Memory

VSIMSA: Configuration file changed: `/home/runner/library.cfg'ALIB: Library "work" attached.work = ./work/work.libMESSAGE "ASRT_7002: Due to optimizations some assertions from source code will be evaluated in non LRM complaint mode. Use switch -ao_off to turn off this optimization."MESSAGE "Unit top modules: tb_data_memory."SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 4 (36.36%) other processes in SLP

SLP: 8 (6.20%) signals in SLP and 8 (6.20%) interface signals

ELAB2: Elaboration final pass complete - time: 0.2 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6422 kB (elbread=459 elab2=5950 kernel=12 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL: Starting Data Memory Verification

KERNEL: WRITE: MEM[1] = 100

KERNEL: PASS : Read/Write Test

KERNEL:

KERNEL: TC2 : Reset Verification

KERNEL: PASS : Reset Cleared Memory

KERNEL:

KERNEL: TC3 : Read Disable Verification

KERNEL: WRITE: MEM[2] = 200

KERNEL: PASS : Read Disable

KERNEL:

KERNEL: TC4 : Multiple Address Verification

KERNEL: WRITE: MEM[1] = 111

KERNEL: WRITE: MEM[2] = 222

KERNEL: WRITE: MEM[3] = 333

KERNEL: PASS : Address 4

KERNEL: PASS : Address 8

KERNEL: PASS : Address 12

KERNEL:

KERNEL: TC5 : Boundary Address Verification

KERNEL: WRITE: MEM[fff] = 9999

KERNEL: PASS : Boundary Address

KERNEL:

KERNEL: TC6 : Mid Address Coverage

KERNEL: WRITE: MEM[400] = 555

KERNEL: PASS : Mid Address

KERNEL:

KERNEL: TC7 : High Address Coverage

KERNEL: WRITE: MEM[800] = 777

KERNEL: PASS : High Address

KERNEL:

KERNEL: TC8 : Read + Write Simultaneously

KERNEL: WRITE: MEM[4] = 1234

KERNEL:

KERNEL: TC9 : Cross Coverage Completion

KERNEL:

KERNEL: TC10 : Low Address + Write OFF

KERNEL:

KERNEL: TC11 : Mid Address Bin Completion

KERNEL: WRITE: MEM[40] = 444

KERNEL: PASS : Mid Address Bin

KERNEL:

KERNEL: Coverage = 100.00%

RUNTIME: Info: RUNTIME_0068 testbench.sv (495): $finish called.

KERNEL: Time: 280 ns, Iteration: 0, Instance: /tb_data_memory, Process: @INITIAL#178_1@.

KERNEL: stopped at time: 280 ns

VSIM: Simulation has finished. There are no more test vectors to simulate.

acdb saveacdb report -db fcover.acdb -txt -o cov.txt -verbose

ACDB: Coverage report "cov.txt" was generated successfully.

exec cat cov.txt

++

++++++++++ REPORT INFO ++++++++++

++

SUMMARY

==

| Property | Value |

==

| User | runner |

| Host | 6c5327bd367a |

| Tool | Riviera-PRO 2025.04 |

```
| Report file      | /home/runner/cov.txt     |
| Report date      | 2026-06-17 07:15         |
| Report arguments | -verbose                 |
| Input file       | /home/runner/fcover.acdb |
| Input file date  | 2026-06-17 07:15         |
| Number of tests  | 1                        |
==================================================

TEST DETAILS
================================================
| Property |              Value              |
================================================
| Test     | fcover.acdb                 |
| Status   | Ok                          |
| Args     | asim +access+r              |
```

| Simtime | 280000 ps |

| Cputime | 0.087 s |

| Seed | 1 |

| Date | 2026-06-17 07:15 |

| User | runner |

| Host | 6c5327bd367a |

| Host os | Linux64 |

| Tool | Riviera-PRO 2025.04 (simulator) |

==

++

++++++++++ DESIGN HIERARCHY ++++++++++

++

CUMULATIVE SUMMARY

===

| Coverage Type | Weight | Hits/Total |

===

| Covergroup Coverage | 1 | 100.000% |

|---------------------|--------|------------|

| Types | | 1 / 1 |

===

CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%

COVERED INSTANCES: 1 / 1

FILES: 1

INSTANCE - /tb_data_memory : work.tb_data_memory

SUMMARY

==

| Coverage Type | Weight | Local Hits/Total | Recursive Hits/Total |

==

| Covergroup Coverage | 1 | 100.000% | 100.000% |

|---------------------|--------|------------------|----------------------|

| Types | | 1 / 1 | 1 / 1 |

==

WEIGHTED AVERAGE LOCAL: 100.000%

WEIGHTED AVERAGE RECURSIVE: 100.000%

COVERGROUP COVERAGE

==

| Covergroup | Hits | Goal / | Status |

```
|                                    |          | At Least |         |
=====================================================================
| TYPE /tb_data_memory/mem_cg        | 100.000% | 100.000% | Covered |
=====================================================================
| INSTANCE                 | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| COVERPOINT ::cp_rst      | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin asserted                       |        2 |        1 | Covered |
| bin deasserted                     |       26 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| COVERPOINT ::cp_read     | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin off                            |       11 |        1 | Covered |
| bin on                             |       17 |        1 | Covered |
|----------------------------------|----------|----------|---------|
```

| COVERPOINT ::cp_write | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| bin off | 18 | 1 | Covered |

| bin on | 10 | 1 | Covered |

|-----------------------------------|----------|----------|---------|

| COVERPOINT ::cp_addr | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| bin low_addr | 17 | 1 | Covered |

| bin mid_addr | 2 | 1 | Covered |

| bin high_addr | 5 | 1 | Covered |

| bin boundary | 4 | 1 | Covered |

|-----------------------------------|----------|----------|---------|

| COVERPOINT ::cp_data | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| bin low_data | 11 | 1 | Covered |

| bin mid_data | 9 | 1 | Covered |

```
| bin high_data                    |        8 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| CROSS ::cross_rw         | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin <off,off>                    |        6 |        1 | Covered |
| bin <off,on>                     |        5 |        1 | Covered |
| bin <on,off>                     |       12 |        1 | Covered |
| bin <on,on>                      |        5 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| CROSS ::cross_addr_write | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin <low_addr,off>               |       11 |        1 | Covered |
| bin <low_addr,on>                |        6 |        1 | Covered |
| bin <mid_addr,off>               |        1 |        1 | Covered |
| bin <mid_addr,on>                |        1 |        1 | Covered |
| bin <high_addr,off>              |        3 |        1 | Covered |
```

```
| bin <high_addr,on>        |     2 |     1 | Covered |

| bin <boundary,off>        |     3 |     1 | Covered |

| bin <boundary,on>         |     1 |     1 | Covered |

=================================================================




++++++++++++++++++++++++++++++++++++++++++++

++++++++++    DESIGN UNITS    ++++++++++

++++++++++++++++++++++++++++++++++++++++++++




CUMULATIVE SUMMARY

=============================================

|   Coverage Type    | Weight | Hits/Total |

=============================================

| Covergroup Coverage |     1 |   100.000% |
```

|---------------------|--------|------------|

| Types | | 1 / 1 |

==

CUMULATIVE DESIGN-BASED COVERAGE: 100.000%

COVERED DESIGN UNITS: 1 / 1

FILES: 1

MODULE - work.tb_data_memory

SUMMARY

==

| Coverage Type | Weight | Hits/Total |

==

| Covergroup Coverage | 1 | 100.000% |

|---------------------|--------|------------|

| Types | | 1 / 1 |

===

WEIGHTED AVERAGE: 100.000%

COVERGROUP COVERAGE

==

| Covergroup | Hits | Goal / | Status |

| | | At Least | |

==

| TYPE /tb_data_memory/mem_cg | 100.000% | 100.000% | Covered |

==

| INSTANCE | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| COVERPOINT ::cp_rst | 100.000% | 100.000% | Covered |

```
|----------------------------------|----------|----------|---------|
| bin asserted                     |        2 |        1 | Covered |
| bin deasserted                   |       26 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| COVERPOINT ::cp_read             | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin off                          |       11 |        1 | Covered |
| bin on                           |       17 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| COVERPOINT ::cp_write            | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
| bin off                          |       18 |        1 | Covered |
| bin on                           |       10 |        1 | Covered |
|----------------------------------|----------|----------|---------|
| COVERPOINT ::cp_addr             | 100.000% | 100.000% | Covered |
|----------------------------------|----------|----------|---------|
```

| bin low_addr | 17 | 1 | Covered |

| bin mid_addr | 2 | 1 | Covered |

| bin high_addr | 5 | 1 | Covered |

| bin boundary | 4 | 1 | Covered |

|-----------------------------------|----------|----------|---------|

| COVERPOINT ::cp_data | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| bin low_data | 11 | 1 | Covered |

| bin mid_data | 9 | 1 | Covered |

| bin high_data | 8 | 1 | Covered |

|-----------------------------------|----------|----------|---------|

| CROSS ::cross_rw | 100.000% | 100.000% | Covered |

|-----------------------------------|----------|----------|---------|

| bin <off,off> | 6 | 1 | Covered |

| bin <off,on> | 5 | 1 | Covered |

| bin <on,off> | 12 | 1 | Covered |

```
| bin <on,on>                     |        5 |        1 | Covered |

|----------------------------------|----------|----------|---------|

| CROSS ::cross_addr_write | 100.000% | 100.000% | Covered |

|----------------------------------|----------|----------|---------|

| bin <low_addr,off>              |       11 |        1 | Covered |

| bin <low_addr,on>               |        6 |        1 | Covered |

| bin <mid_addr,off>              |        1 |        1 | Covered |

| bin <mid_addr,on>               |        1 |        1 | Covered |

| bin <high_addr,off>             |        3 |        1 | Covered |

| bin <high_addr,on>              |        2 |        1 | Covered |

| bin <boundary,off>              |        3 |        1 | Covered |

| bin <boundary,on>               |        1 |        1 | Covered |
exit
KERNEL:
KERNEL: Final Coverage = 100.00%
VSIM: Simulation has finished.
```

8. EX_MEM

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_ex_mem."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 2 (50.00%) other processes in SLP

SLP: 20 (12.50%) signals in SLP and 20 (12.50%) interface signals

ELAB2: Elaboration final pass complete - time: 0.2 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5360 kB (elbread=459 elab2=4888 kernel=13 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : Reset Verification

KERNEL: PASS : Reset Verification

KERNEL:

KERNEL: TC2 : Data Path Transfer

KERNEL: PASS : Data Path Transfer

KERNEL: ASSERT PASS : DATA PATH

KERNEL:

KERNEL: TC3 : Control Signal Transfer

KERNEL: PASS : Control Signal Transfer

KERNEL: ASSERT PASS : CONTROL PATH

KERNEL:

KERNEL: TC4 : PC+4 Transfer

KERNEL: PASS : PC+4 Transfer

KERNEL: ASSERT PASS : PC+4

KERNEL:

KERNEL: TC5 : result_src Transfer

KERNEL: PASS : result_src Transfer

KERNEL: ASSERT PASS : RESULT_SRC

KERNEL:

KERNEL: TC6 : result_src = MEM

KERNEL: ASSERT PASS : RESULT_SRC_MEM

KERNEL:

KERNEL: TC7 : result_src = DEFAULT

KERNEL: ASSERT PASS : RESULT_SRC_DEFAULT

```
# KERNEL:
# KERNEL: TC8 : Mid Register
# KERNEL: ASSERT PASS : MID_REGISTER
# KERNEL:
# KERNEL: TC9 : High Register
# KERNEL: ASSERT PASS : HIGH_REGISTER
# KERNEL: ASSERT PASS : MEM_PATH_REGWRITE_ON
# KERNEL: ASSERT PASS : MEM_PATH_REGWRITE_OFF
# KERNEL: ASSERT PASS : DEFAULT_PATH_REGWRITE_ON
# KERNEL: ASSERT PASS : DEFAULT_PATH_REGWRITE_OFF
# KERNEL:
# KERNEL: TC14 : PC4 Path + RegWrite OFF
# KERNEL: ASSERT PASS : PC4_PATH_REGWRITE_OFF
# KERNEL:
# KERNEL: EX_MEM VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (430): $finish called.
# KERNEL: Time: 152 ns,  Iteration: 0,  Instance: /tb_ex_mem,  Process: @INITIAL#136_1@.
# KERNEL: stopped at time: 152 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
```

```
# |     Property     |          Value          |
# ==================================================
# | User             | runner                  |
# | Host             | ad2cb349f15b            |
# | Tool             | Riviera-PRO 2025.04     |
# | Report file      | /home/runner/cov.txt    |
# | Report date      | 2026-06-17 10:35        |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-17 10:35        |
# | Number of tests  | 1                       |
# ==================================================
#
#
# TEST DETAILS
# ================================================
# | Property |             Value             |
# ================================================
# | Test     | fcover.acdb:fcover            |
# | Status   | Ok                            |
# | Args     | asim +access+r                |
# | Simtime  | 152000 ps                     |
# | Cputime  | 0.186 s                       |
# | Seed     | 1                             |
# | Date     | 2026-06-17 10:35              |
# | User     | runner                        |
# | Host     | ad2cb349f15b                  |
# | Host os  | Linux64                       |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
```

```
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_ex_mem : work.tb_ex_mem
#
#
#    SUMMARY
#
========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
```

```
#
==========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
==========================================================================
========
#    |                Covergroup                |   Hits   |  Goal /  | Status  |
#    |                                          |          | At Least |         |
#
==========================================================================
========
#    | TYPE /tb_ex_mem/ex_mem_cg                | 100.000% | 100.000% | Covered |
#
==========================================================================
========
#    | INSTANCE EX_MEM_COVERAGE                 | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_rst       | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin asserted                             |        1 |        1 | Covered |
#    | bin deasserted                           |       14 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_mem_read  | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin off                                  |        2 |        1 | Covered |
#    | bin on                                   |       13 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_mem_write | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin off                                  |        2 |        1 | Covered |
```

```
#     | bin on                                    |       13 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_mem_to_reg    | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin off                                   |        2 |        1 | Covered |
#     | bin on                                    |       13 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_reg_write     | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin off                                   |        6 |        1 | Covered |
#     | bin on                                    |        9 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_result_src    | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin alu_path                                  |        4 |        1 | Covered |
#     | bin mem_path                                  |        3 |        1 | Covered |
#     | bin pc4_path                                  |        3 |        1 | Covered |
#     | bin default_path                              |        5 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_rd            | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin x0                                    |        1 |        1 | Covered |
#     | bin low_reg                                   |        6 |        1 | Covered |
#     | bin mid_reg                                   |        1 |        1 | Covered |
#     | bin high_reg                                  |        7 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | CROSS EX_MEM_COVERAGE::cross_result_regwrite | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin <alu_path,off>                            |        2 |        1 | Covered |
#     | bin <alu_path,on>                             |        2 |        1 | Covered |
#     | bin <mem_path,off>                            |        1 |        1 | Covered |
#     | bin <mem_path,on>                             |        2 |        1 | Covered |
```

```
#     | bin <pc4_path,off>                        |        2 |        1 | Covered |
#     | bin <pc4_path,on>                         |        1 |        1 | Covered |
#     | bin <default_path,off>                    |        1 |        1 | Covered |
#     | bin <default_path,on>                     |        4 |        1 | Covered |
#
========================================================================
========
#
#
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_ex_mem
#
#
#     SUMMARY
#     =============================================
#     |    Coverage Type    | Weight | Hits/Total |
```

```
#     ================================================
#     | Covergroup Coverage |       1 |   100.000% |
#     |---------------------|---------|------------|
#     | Types               |         |      1 / 1 |
#     ================================================
#     WEIGHTED AVERAGE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
==============================================================================
#     |                 Covergroup                |   Hits   |  Goal /  | Status  |
#     |                                           |          | At Least |         |
#
==============================================================================
#     | TYPE /tb_ex_mem/ex_mem_cg                 | 100.000% | 100.000% | Covered |
#
==============================================================================
#     | INSTANCE EX_MEM_COVERAGE                  | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_rst        | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin asserted                              |        1 |        1 | Covered |
#     | bin deasserted                            |       14 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_mem_read   | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin off                                   |        2 |        1 | Covered |
#     | bin on                                    |       13 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT EX_MEM_COVERAGE::cp_mem_write  | 100.000% | 100.000% | Covered |
```

```
#    |--------------------------------------------|----------|----------|---------|
#    | bin off                                   |        2 |         1 | Covered |
#    | bin on                                    |       13 |         1 | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_mem_to_reg    | 100.000% | 100.000% | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | bin off                                   |        2 |         1 | Covered |
#    | bin on                                    |       13 |         1 | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_reg_write     | 100.000% | 100.000% | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | bin off                                   |        6 |         1 | Covered |
#    | bin on                                    |        9 |         1 | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_result_src    | 100.000% | 100.000% | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | bin alu_path                              |        4 |         1 | Covered |
#    | bin mem_path                              |        3 |         1 | Covered |
#    | bin pc4_path                              |        3 |         1 | Covered |
#    | bin default_path                          |        5 |         1 | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | COVERPOINT EX_MEM_COVERAGE::cp_rd            | 100.000% | 100.000% | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | bin x0                                    |        1 |         1 | Covered |
#    | bin low_reg                               |        6 |         1 | Covered |
#    | bin mid_reg                               |        1 |         1 | Covered |
#    | bin high_reg                              |        7 |         1 | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | CROSS EX_MEM_COVERAGE::cross_result_regwrite | 100.000% | 100.000% | Covered |
#    |--------------------------------------------|----------|----------|---------|
#    | bin <alu_path,off>                        |        2 |         1 | Covered |
#    | bin <alu_path,on>                         |        2 |         1 | Covered |
```

```
#     | bin <mem_path,off>                              |         1 |         1 | Covered |
#     | bin <mem_path,on>                               |         2 |         1 | Covered |
#     | bin <pc4_path,off>                              |         2 |         1 | Covered |
#     | bin <pc4_path,on>                               |         1 |         1 | Covered |
#     | bin <default_path,off>                          |         1 |         1 | Covered |
#     | bin <default_path,on>                           |         4 |         1 | Covered |
#
================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

9. FORWARDING

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_forwarding."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 8 (13.79%) signals in SLP and 8 (13.79%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5313 kB (elbread=459 elab2=4844 kernel=9 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : NO FORWARDING

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : NO FORWARDING

KERNEL:

KERNEL: TC2 : EX FORWARD A

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : EX FORWARD A

KERNEL:

KERNEL: TC3 : EX FORWARD B

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : EX FORWARD B

KERNEL:

KERNEL: TC4 : MEM FORWARD A

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : MEM FORWARD A

KERNEL:

KERNEL: TC5 : MEM FORWARD B

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : MEM FORWARD B

KERNEL:

KERNEL: TC6 : PRIORITY CHECK A

KERNEL: PASS : TC6

KERNEL: ASSERT PASS : EX PRIORITY

KERNEL:

```
# KERNEL: TC7 : PRIORITY CHECK B
# KERNEL: PASS : TC7
# KERNEL: ASSERT PASS : EX PRIORITY
# KERNEL:
# KERNEL: TC8 : ZERO REGISTER
# KERNEL: PASS : TC8
# KERNEL: ASSERT PASS : ZERO REGISTER
# KERNEL:
# KERNEL: FORWARDING VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (261): $finish called.
# KERNEL: Time: 50 ns,  Iteration: 0,  Instance: /tb_forwarding,  Process: @INITIAL#76_0@.
# KERNEL: stopped at time: 50 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | f81a773cb80b             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
```

```
# | Report date      | 2026-06-19 04:59          |
# | Report arguments | -verbose                  |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-19 04:59          |
# | Number of tests  | 1                         |
# ==================================================
#
#
# TEST DETAILS
# ================================================
# | Property |              Value              |
# ================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 50000 ps                        |
# | Cputime  | 0.024 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-19 04:59                |
# | User     | runner                          |
# | Host     | f81a773cb80b                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
```

```
# ==================================================
# |     Coverage Type     | Weight | Hits/Total |
# ==================================================
# | Covergroup Coverage |       1 |    100.000% |
# |---------------------|--------|------------|
# | Types               |        |       1 / 1 |
# ==================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_forwarding : work.tb_forwarding
#
#
#     SUMMARY
#
==============================================================================
==
#     |     Coverage Type     | Weight | Local Hits/Total | Recursive Hits/Total |
#
==============================================================================
==
#     | Covergroup Coverage |       1 |          100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
==============================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
```

```
# ============================================================================================
#     |                 Covergroup                 |   Hits   |  Goal /  | Status  |
#     |                                            |          | At Least |         |
# ============================================================================================
#     | TYPE /tb_forwarding/forwarding_cg          | 100.000% | 100.000% | Covered |
# ============================================================================================
#     | INSTANCE FORWARDING_COVERAGE               | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT FORWARDING_COVERAGE::cp_forward_A | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin no_forward                             |        5 |        1 | Covered |
#     | bin ex_forward                             |        2 |        1 | Covered |
#     | bin mem_forward                            |        1 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT FORWARDING_COVERAGE::cp_forward_B | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin no_forward                             |        5 |        1 | Covered |
#     | bin ex_forward                             |        2 |        1 | Covered |
#     | bin mem_forward                            |        1 |        1 | Covered |
# ============================================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++++       DESIGN UNITS       ++++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++++++++++
#
```

```
#
# CUMULATIVE SUMMARY
# =================================================
# |    Coverage Type    | Weight | Hits/Total |
# =================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_forwarding
#
#
#    SUMMARY
#    =================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    =================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    =================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
=============================================================================
========
#    |                Covergroup                | Hits   | Goal /  | Status  |
```

```
#     |                                  |        | At Least |        |
#
================================================================================
#     | TYPE /tb_forwarding/forwarding_cg          | 100.000% | 100.000% | Covered |
#
================================================================================
#     | INSTANCE FORWARDING_COVERAGE               | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT FORWARDING_COVERAGE::cp_forward_A | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin no_forward                             |        5 |        1 | Covered |
#     | bin ex_forward                             |        2 |        1 | Covered |
#     | bin mem_forward                            |        1 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT FORWARDING_COVERAGE::cp_forward_B | 100.000% | 100.000% |
Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin no_forward                             |        5 |        1 | Covered |
#     | bin ex_forward                             |        2 |        1 | Covered |
#     | bin mem_forward                            |        1 |        1 | Covered |
#
================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

10.Hazard Detection

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_hazard_detection."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.0 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (20.00%) other processes in SLP

SLP: 8 (13.79%) signals in SLP and 8 (13.79%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5314 kB (elbread=459 elab2=4845 kernel=9 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : NO HAZARD

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : NO HAZARD

KERNEL:

KERNEL: TC2 : RS1 HAZARD

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : RS1 HAZARD

KERNEL:

KERNEL: TC3 : RS2 HAZARD

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : RS2 HAZARD

KERNEL:

KERNEL: TC4 : MEMREAD DISABLED

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : MEMREAD DISABLED

KERNEL:

KERNEL: TC5 : ZERO REGISTER

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : ZERO REGISTER

KERNEL:

KERNEL: TC6 : DOUBLE HAZARD

KERNEL: PASS : TC6

KERNEL: ASSERT PASS : DOUBLE HAZARD

KERNEL:

KERNEL: TC7 : DYNAMIC SWITCHING

KERNEL: ASSERT PASS : NORMAL

```
# KERNEL: ASSERT PASS : HAZARD
# KERNEL: ASSERT PASS : RECOVERY
# KERNEL:
# KERNEL: HAZARD DETECTION VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (237): $finish called.
# KERNEL: Time: 55 ns,  Iteration: 0,  Instance: /tb_hazard_detection,  Process: @INITIAL#58_3@.
# KERNEL: stopped at time: 55 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | ac32f0c33b04             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-19 05:07         |
# | Report arguments | -verbose                 |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-19 05:07         |
# | Number of tests  | 1                        |
```

```
# ====================================================
#
#
# TEST DETAILS
# ===================================================
# | Property |              Value              |
# ===================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 55000 ps                        |
# | Cputime  | 0.002 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-19 05:07                |
# | User     | runner                          |
# | Host     | ac32f0c33b04                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ===================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
```

```
# | Types                 |        |      1 / 1 |
# ==================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_hazard_detection : work.tb_hazard_detection
#
#
#     SUMMARY
#
================================================================================
==
#     |    Coverage Type     | Weight | Local Hits/Total | Recursive Hits/Total |
#
================================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
================================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
================================================================================
======
#     |                  Covergroup                  |   Hits   |  Goal /  | Status  |
#     |                                              |          | At Least |         |
```

```
#
==============================================================================
======
#    | TYPE /tb_hazard_detection/hazard_cg        | 100.000% | 100.000% | Covered |
#
==============================================================================
======
#    | INSTANCE HAZARD_COVERAGE                   | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT HAZARD_COVERAGE::cp_pc_write    | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin stall                                |        3 |        1 | Covered |
#    | bin normal                               |        4 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT HAZARD_COVERAGE::cp_control_mux | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin nop_inserted                         |        3 |        1 | Covered |
#    | bin normal_flow                          |        4 |        1 | Covered |
#
==============================================================================
======
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ============================================
# |    Coverage Type    | Weight | Hits/Total |
# ============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
```

```
# | Types              |       |      1 / 1 |
# ================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_hazard_detection
#
#
#    SUMMARY
#    ================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
=============================================================================
======
#    |                Covergroup                |  Hits   |  Goal /  | Status  |
#    |                                          |         | At Least |         |
#
=============================================================================
======
#    | TYPE /tb_hazard_detection/hazard_cg       | 100.000% | 100.000% | Covered |
#
=============================================================================
======
```

```
#     | INSTANCE HAZARD_COVERAGE                  | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT HAZARD_COVERAGE::cp_pc_write    | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin stall                                |        3 |        1 | Covered |
#     | bin normal                               |        4 |        1 | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT HAZARD_COVERAGE::cp_control_mux | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin nop_inserted                         |        3 |        1 | Covered |
#     | bin normal_flow                          |        4 |        1 | Covered |
#
================================================================================
======
#
exit
# VSIM: Simulation has finished
```

11.ID_EX

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_ID_EX."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 3 (37.50%) other processes in SLP

SLP: 44 (25.14%) signals in SLP and 42 (24.00%) interface signals

ELAB2: Elaboration final pass complete - time: 0.5 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6388 kB (elbread=459 elab2=5917 kernel=11 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC1

KERNEL:

KERNEL: TC2 : NORMAL TRANSFER

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: PASS : TC2

KERNEL:

KERNEL: TC3 : STALL

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : TRANSFER

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC3

KERNEL:

KERNEL: TC4 : FLUSH

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC4

KERNEL:

KERNEL: TC5 : FLUSH PRIORITY

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC5

KERNEL:

KERNEL: TC6 : MEM PATH

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : STALL

KERNEL: PASS : TC6

KERNEL:

KERNEL: TC7 : PC+4 PATH

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC7

KERNEL:

KERNEL: TC8 : SECOND TRANSFER

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : TRANSFER

KERNEL: PASS : TC8

KERNEL:

KERNEL: ID_EX VERIFICATION COMPLETE

KERNEL: ASSERT PASS : RESET

KERNEL: ASSERT PASS : FLUSH

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : TRANSFER

RUNTIME: Info: RUNTIME_0068 testbench.sv (571): $finish called.

KERNEL: Time: 92 ns, Iteration: 0, Instance: /tb_ID_EX, Process: @INITIAL#305_1@.

KERNEL: stopped at time: 92 ns

VSIM: Simulation has finished. There are no more test vectors to simulate.

acdb save

acdb report -db fcover.acdb -txt -o cov.txt -verbose

ACDB: Coverage report "cov.txt" was generated successfully.

exec cat cov.txt

+++

++++++++++   REPORT INFO   ++++++++++

+++

#

#

SUMMARY

===

| Property | Value |

===

| User | runner |

| Host | 4ce0396af2e9 |

| Tool | Riviera-PRO 2025.04 |

| Report file | /home/runner/cov.txt |

| Report date | 2026-06-19 04:50 |

| Report arguments | -verbose |

| Input file | /home/runner/fcover.acdb |

| Input file date | 2026-06-19 04:50 |

| Number of tests | 1 |

===

#

```
#
# TEST DETAILS
# ================================================
# | Property |              Value              |
# ================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 92000 ps                        |
# | Cputime  | 0.128 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-19 04:50                |
# | User     | runner                          |
# | Host     | 4ce0396af2e9                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
```

```
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_ID_EX : work.tb_ID_EX
#
#
#     SUMMARY
#
==========================================================================
==
#     |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
==========================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
==========================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
==========================================================================
====
#     |                Covergroup                |   Hits   |  Goal /  | Status  |
#     |                                          |          | At Least |         |
#
==========================================================================
====
#     | TYPE /tb_ID_EX/id_ex_cg                  | 100.000% | 100.000% | Covered |
```

```
#
=========================================================================
====
#     | INSTANCE ID_EX_COVERAGE                | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_rst       | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin reset_on                            |        1 |        1 | Covered |
#     | bin reset_off                           |        8 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_flush     | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin flush_on                            |        2 |        1 | Covered |
#     | bin flush_off                           |        7 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_stall     | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin stall_on                            |        2 |        1 | Covered |
#     | bin stall_off                           |        7 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_regwrite  | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin enabled                             |        4 |        1 | Covered |
#     | bin disabled                            |        4 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_result_src | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin alu_path                            |        5 |        1 | Covered |
#     | bin mem_path                            |        2 |        1 | Covered |
#     | bin pc4_path                            |        1 |        1 | Covered |
#
=========================================================================
====
#
```

```
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN UNITS     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_ID_EX
#
#
#    SUMMARY
#    =============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    =============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    =============================================
#    WEIGHTED AVERAGE: 100.000%
#
```

```
#
#     COVERGROUP COVERAGE
#
==========================================================================
====
#     |                 Covergroup                |   Hits   |  Goal /  | Status  |
#     |                                           |          | At Least |         |
#
==========================================================================
====
#     | TYPE /tb_ID_EX/id_ex_cg                   | 100.000% | 100.000% | Covered |
#
==========================================================================
====
#     | INSTANCE ID_EX_COVERAGE                   | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_rst         | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin reset_on                              |        1 |        1 | Covered |
#     | bin reset_off                             |        8 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_flush       | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin flush_on                              |        2 |        1 | Covered |
#     | bin flush_off                             |        7 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_stall       | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin stall_on                              |        2 |        1 | Covered |
#     | bin stall_off                             |        7 |        1 | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_regwrite    | 100.000% | 100.000% | Covered |
#     |-------------------------------------------|----------|----------|---------|
#     | bin enabled                               |        4 |        1 | Covered |
```

```
#     | bin disabled                              |        4 |         1 | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT ID_EX_COVERAGE::cp_result_src | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin alu_path                              |        5 |         1 | Covered |
#     | bin mem_path                              |        2 |         1 | Covered |
#     | bin pc4_path                              |        1 |         1 | Covered |
#
==============================================================================
====
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

12.IF_ID

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_IF_ID."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 2 (25.00%) other processes in SLP

SLP: 10 (12.05%) signals in SLP and 10 (12.05%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6357 kB (elbread=459 elab2=5887 kernel=10 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : RESET

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : RESET

KERNEL:

KERNEL: TC2 : NORMAL TRANSFER

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : TRANSFER

KERNEL:

KERNEL: TC3 : STALL

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : STALL

KERNEL:

KERNEL: TC4 : FLUSH

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : FLUSH

KERNEL:

KERNEL: TC5 : FLUSH PRIORITY

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : FLUSH PRIORITY

KERNEL:

KERNEL: TC6 : NORMAL AGAIN

KERNEL: PASS : TC6

KERNEL: ASSERT PASS : NORMAL AGAIN

KERNEL:

```
# KERNEL: IF_ID VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (351): $finish called.
# KERNEL: Time: 72 ns,  Iteration: 0,  Instance: /tb_IF_ID,  Process: @INITIAL#142_1@.
# KERNEL: stopped at time: 72 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | 4e17ec9d314f             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-19 04:24         |
# | Report arguments | -verbose                 |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-19 04:24         |
# | Number of tests  | 1                        |
# ===============================================
#
#
```

TEST DETAILS

==

| Property | Value |

==

| Test | fcover.acdb:fcover |

| Status | Ok |

| Args | asim +access+r |

| Simtime | 72000 ps |

| Cputime | 0.087 s |

| Seed | 1 |

| Date | 2026-06-19 04:24 |

| User | runner |

| Host | 4e17ec9d314f |

| Host os | Linux64 |

| Tool | Riviera-PRO 2025.04 (simulator) |

==

#

#

++

++++++++++   DESIGN HIERARCHY   ++++++++++

++

#

#

CUMULATIVE SUMMARY

===

| Coverage Type | Weight | Hits/Total |

===

| Covergroup Coverage | 1 | 100.000% |

|---------------------|--------|------------|

| Types | | 1 / 1 |

===

CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%

```
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_IF_ID : work.tb_IF_ID
#
#
#    SUMMARY
#
============================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
============================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
============================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===========================================================================
#    |            Covergroup            |   Hits   |  Goal /  | Status  |
#    |                                  |          | At Least |         |
#
===========================================================================
#    | TYPE /tb_IF_ID/ifid_cg           | 100.000% | 100.000% | Covered |
#
===========================================================================
```

```
#     | INSTANCE IF_ID_COVERAGE              | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_rst   | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | bin reset_on                      |        1 |        1 | Covered |
#     | bin reset_off                     |        6 |        1 | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_flush | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | bin flush_on                      |        2 |        1 | Covered |
#     | bin flush_off                     |        5 |        1 | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_write | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | bin write_on                      |        4 |        1 | Covered |
#     | bin write_off                     |        3 |        1 | Covered |
#
================================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ============================================
# |    Coverage Type    | Weight | Hits/Total |
# ============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ============================================
```

```
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_IF_ID
#
#
#     SUMMARY
#     ================================================
#     |    Coverage Type    | Weight | Hits/Total |
#     ================================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |---------------------|--------|------------|
#     | Types               |        |      1 / 1 |
#     ================================================
#     WEIGHTED AVERAGE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
=============================================================================
#     |            Covergroup              |   Hits   |  Goal /  | Status  |
#     |                                    |          | At Least |         |
#
=============================================================================
#     | TYPE /tb_IF_ID/ifid_cg             | 100.000% | 100.000% | Covered |
#
=============================================================================
#     | INSTANCE IF_ID_COVERAGE            | 100.000% | 100.000% | Covered |
#     |------------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_rst  | 100.000% | 100.000% | Covered |
#     |------------------------------------|----------|----------|---------|
```

```
#     | bin reset_on                        |        1 |        1 | Covered |
#     | bin reset_off                       |        6 |        1 | Covered |
#     |------------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_flush | 100.000% | 100.000% | Covered |
#     |------------------------------------|----------|----------|---------|
#     | bin flush_on                        |        2 |        1 | Covered |
#     | bin flush_off                       |        5 |        1 | Covered |
#     |------------------------------------|----------|----------|---------|
#     | COVERPOINT IF_ID_COVERAGE::cp_write | 100.000% | 100.000% | Covered |
#     |------------------------------------|----------|----------|---------|
#     | bin write_on                        |        4 |        1 | Covered |
#     | bin write_off                       |        3 |        1 | Covered |
#
================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

13.imm_gen

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_imm_gen."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 2 (4.35%) signals in SLP and 2 (4.35%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5315 kB (elbread=459 elab2=4845 kernel=10 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : I-TYPE

KERNEL: ASSERT PASS : I_TYPE

KERNEL: IMM = 100

KERNEL:

KERNEL: TC2 : LOAD

KERNEL: ASSERT PASS : LOAD

KERNEL: IMM = 32

KERNEL:

KERNEL: TC3 : JALR

KERNEL: ASSERT PASS : JALR

KERNEL: IMM = 16

KERNEL:

KERNEL: TC4 : STORE

KERNEL: ASSERT PASS : STORE

KERNEL: IMM = 100

KERNEL:

KERNEL: TC5 : BRANCH

KERNEL: ASSERT PASS : BRANCH

KERNEL: IMM = 16

KERNEL:

KERNEL: TC6 : LUI

KERNEL: ASSERT PASS : LUI

KERNEL: IMM = 305418240

KERNEL:

```
# KERNEL: TC7 : AUIPC
# KERNEL: ASSERT PASS : AUIPC
# KERNEL: IMM = 305418240
# KERNEL:
# KERNEL: TC8 : JAL
# KERNEL: ASSERT PASS : JAL
# KERNEL: IMM = 32
# KERNEL:
# KERNEL: TC9 : INVALID
# KERNEL: ASSERT PASS : INVALID OPCODE
# KERNEL: IMM = 0
# KERNEL:
# KERNEL: TC10 : ZERO IMM
# KERNEL: ASSERT PASS : ZERO IMM
# KERNEL: IMM = 0
# KERNEL:
# KERNEL: TC11 : MEDIUM IMM
# KERNEL: ASSERT PASS : MEDIUM IMM
# KERNEL: IMM = 500
# KERNEL:
# KERNEL: TC12 : FORCE MEDIUM BIN
# KERNEL: ASSERT PASS : MEDIUM IMM
# KERNEL: IMM = 800
# KERNEL:
# KERNEL: IMM_GEN VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (279): $finish called.
# KERNEL: Time: 75 ns,  Iteration: 0,  Instance: /tb_imm_gen,  Process: @INITIAL#63_0@.
# KERNEL: stopped at time: 75 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
```

```
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++      REPORT INFO      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# =================================================
# |    Property    |         Value          |
# =================================================
# | User            | runner                 |
# | Host            | 4c59741f8ddb           |
# | Tool            | Riviera-PRO 2025.04    |
# | Report file     | /home/runner/cov.txt   |
# | Report date     | 2026-06-18 06:37       |
# | Report arguments | -verbose              |
# | Input file      | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 06:37      |
# | Number of tests  | 1                     |
# =================================================
#
#
# TEST DETAILS
# ================================================
# | Property |            Value            |
# ================================================
# | Test     | fcover.acdb:fcover          |
# | Status   | Ok                          |
# | Args     | asim +access+r              |
# | Simtime  | 75000 ps                    |
```

```
# | Cputime  | 0.021 s                    |
# | Seed     | 1                          |
# | Date     | 2026-06-18 06:37           |
# | User     | runner                     |
# | Host     | 4c59741f8ddb               |
# | Host os  | Linux64                    |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_imm_gen : work.tb_imm_gen
#
#
#    SUMMARY
```

```
#
================================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
================================================================================
==
#    | Covergroup Coverage |      1 |        100.000% |            100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |           1 / 1 |               1 / 1 |
#
================================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
================================================================================
===
#    |               Covergroup               |   Hits   |  Goal /  |  Status  |
#    |                                        |          | At Least |          |
#
================================================================================
===
#    | TYPE /tb_imm_gen/imm_gen_cg            | 100.000% | 100.000% | Covered  |
#
================================================================================
===
#    | INSTANCE IMM_GEN_COVERAGE              | 100.000% | 100.000% | Covered  |
#    |----------------------------------------|----------|----------|----------|
#    | COVERPOINT IMM_GEN_COVERAGE::cp_opcode | 100.000% | 100.000% | Covered  |
#    |----------------------------------------|----------|----------|----------|
#    | bin i_type                             |        4 |        1 | Covered  |
#    | bin load                               |        1 |        1 | Covered  |
#    | bin jalr                               |        1 |        1 | Covered  |
```

```
#     | bin store                          |        1 |        1 | Covered  |
#     | bin branch                         |        1 |        1 | Covered  |
#     | bin lui                            |        1 |        1 | Covered  |
#     | bin auipc                          |        1 |        1 | Covered  |
#     | bin jal                            |        1 |        1 | Covered  |
#     | default bin invalid                |        1 |    -     | Occurred |
#     |--------------------------------------|----------|----------|----------|
#     | COVERPOINT IMM_GEN_COVERAGE::cp_imm    | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | bin zero_val                       |        2 |        1 | Covered  |
#     | bin small_pos                      |        6 |        1 | Covered  |
#     | bin medium_pos                     |        1 |        1 | Covered  |
#     | bin large_pos                      |        2 |        1 | Covered  |
#
=====================================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ============================================
# |    Coverage Type    | Weight | Hits/Total |
# ============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
```

```
# FILES: 1
#
#
# MODULE - work.tb_imm_gen
#
#
#    SUMMARY
#    ================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===============================================================================
#    |             Covergroup              |  Hits   |  Goal /  |  Status  |
#    |                                     |         | At Least |          |
#
===============================================================================
#    | TYPE /tb_imm_gen/imm_gen_cg          | 100.000% | 100.000% | Covered  |
#
===============================================================================
#    | INSTANCE IMM_GEN_COVERAGE              | 100.000% | 100.000% | Covered  |
#    |----------------------------------------|----------|----------|----------|
#    | COVERPOINT IMM_GEN_COVERAGE::cp_opcode | 100.000% | 100.000% | Covered  |
#    |----------------------------------------|----------|----------|----------|
```

```
#     | bin i_type                          |        4 |        1 | Covered  |
#     | bin load                            |        1 |        1 | Covered  |
#     | bin jalr                            |        1 |        1 | Covered  |
#     | bin store                           |        1 |        1 | Covered  |
#     | bin branch                          |        1 |        1 | Covered  |
#     | bin lui                             |        1 |        1 | Covered  |
#     | bin auipc                           |        1 |        1 | Covered  |
#     | bin jal                             |        1 |        1 | Covered  |
#     | default bin invalid                 |        1 |    -     | Occurred |
#     |--------------------------------------|----------|----------|----------|
#     | COVERPOINT IMM_GEN_COVERAGE::cp_imm    | 100.000% | 100.000% | Covered  |
#     |--------------------------------------|----------|----------|----------|
#     | bin zero_val                        |        2 |        1 | Covered  |
#     | bin small_pos                       |        6 |        1 | Covered  |
#     | bin medium_pos                      |        1 |        1 | Covered  |
#     | bin large_pos                       |        2 |        1 | Covered  |
#
=================================================================================
===
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished
```

14. Instruction Memory

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_instruction_memory."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.0 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 2 (50.00%) other processes in SLP

SLP: 2 (5.88%) signals in SLP and 3 (8.82%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5308 kB (elbread=459 elab2=4840 kernel=8 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : Address 0

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : ADDR0

KERNEL:

KERNEL: TC2 : Address 1

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : ADDR1

KERNEL:

KERNEL: TC3 : Address 2

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : ADDR2

KERNEL:

KERNEL: TC4 : Address 3

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : ADDR3

KERNEL:

KERNEL: TC5 : Alignment Check

KERNEL: PASS : TC5

KERNEL:

KERNEL: TC6 : Alignment Check

KERNEL: PASS : TC6

KERNEL:

KERNEL: TC7 : Alignment Check

KERNEL: PASS : TC7

```
# KERNEL:
# KERNEL: INSTRUCTION_MEMORY VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (178): $finish called.
# KERNEL: Time: 45 ns,  Iteration: 0,  Instance: /tb_instruction_memory,  Process: @INITIAL#40_0@.
# KERNEL: stopped at time: 45 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property     |          Value            |
# ===============================================
# | User            | runner                    |
# | Host            | c6f1187e3dd9              |
# | Tool            | Riviera-PRO 2025.04       |
# | Report file     | /home/runner/cov.txt      |
# | Report date     | 2026-06-18 15:45          |
# | Report arguments | -verbose                 |
# | Input file      | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 15:45         |
# | Number of tests  | 1                        |
# ===============================================
#
```

```
#
# TEST DETAILS
# ==================================================
# | Property |              Value              |
# ==================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 45000 ps                        |
# | Cputime  | 0.032 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-18 15:45                |
# | User     | runner                          |
# | Host     | c6f1187e3dd9                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
```

```
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
# INSTANCE - /tb_instruction_memory : work.tb_instruction_memory
#     SUMMARY
#
=====================================================================
==
#     |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
=====================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
=====================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
=====================================================================
====
#     |                Covergroup                |   Hits   |  Goal /  | Status  |
#     |                                          |          | At Least |         |
#
=====================================================================
====
#     | TYPE /tb_instruction_memory/instr_mem_cg | 100.000% | 100.000% | Covered |
=====================================================================
====
#     | INSTANCE INSTR_MEM_COVERAGE              | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT INSTR_MEM_COVERAGE::cp_pc     | 100.000% | 100.000% | Covered |
```

```
#     |-----------------------------------------|----------|----------|---------|
#     | bin addr0                          |        4 |        1 | Covered |
#     | bin addr1                          |        1 |        1 | Covered |
#     | bin addr2                          |        1 |        1 | Covered |
#     | bin addr3                          |        1 |        1 | Covered |
#
==========================================================================
====
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_instruction_memory
#
#
#     SUMMARY
#     ==============================================
#     |    Coverage Type    | Weight | Hits/Total |
#     ==============================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |---------------------|--------|------------|
```

```
#    | Types              |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
=============================================================================
#    |                Covergroup               |   Hits   |  Goal /  | Status  |
#    |                                         |          | At Least |         |
#
=============================================================================
#    | TYPE /tb_instruction_memory/instr_mem_cg | 100.000% | 100.000% | Covered |
#
=============================================================================
#    | INSTANCE INSTR_MEM_COVERAGE              | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | COVERPOINT INSTR_MEM_COVERAGE::cp_pc     | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | bin addr0                               |        4 |        1 | Covered |
#    | bin addr1                               |        1 |        1 | Covered |
#    | bin addr2                               |        1 |        1 | Covered |
#    | bin addr3                               |        1 |        1 | Covered |
#
=============================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

15. Mem_wb

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_mem_wb."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 2 (50.00%) other processes in SLP

SLP: 16 (12.70%) signals in SLP and 16 (12.70%) interface signals

ELAB2: Elaboration final pass complete - time: 0.2 [s].

KERNEL: SLP loading done - time: 0.0 [s].

```
# KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.
# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.
# KERNEL: SLP simulation initialization done - time: 0.0 [s].
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 5340 kB (elbread=459 elab2=4870 kernel=11 sdf=0)
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
run -all
# KERNEL: PASS : TC1 Reset
# KERNEL: ASSERT PASS : ALU TRANSFER
# KERNEL: PASS : TC2 Data Path
# KERNEL: ASSERT PASS : REG_WRITE
# KERNEL: ASSERT PASS : MEM_TO_REG
# KERNEL: PASS : TC3 Control Path
# KERNEL: ASSERT PASS : PC4
# KERNEL: PASS : TC4 PC+4
# KERNEL: ASSERT PASS : RESULT_SRC
# KERNEL: PASS : TC5 result_src
# KERNEL: ASSERT PASS : RD
# KERNEL: PASS : TC6 Update Test
# KERNEL:
# KERNEL: MEM_WB VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (333): $finish called.
# KERNEL: Time: 142 ns,  Iteration: 0,  Instance: /tb_mem_wb,  Process: @INITIAL#106_1@.
# KERNEL: stopped at time: 142 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
```

```
# ++++++++++      REPORT INFO      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property     |          Value           |
# ===============================================
# | User             | runner                  |
# | Host             | 9562525e2095            |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-17 10:07         |
# | Report arguments | -verbose                 |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-17 10:07         |
# | Number of tests  | 1                        |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |              Value              |
# ==============================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 142000 ps                       |
# | Cputime  | 0.126 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-17 10:07                |
# | User     | runner                          |
```

| Host | 9562525e2095 |
| Host os | Linux64 |
| Tool | Riviera-PRO 2025.04 (simulator) |
===
#
#
++
++++++++++   DESIGN HIERARCHY   ++++++++++
++
#
#
CUMULATIVE SUMMARY
==
| Coverage Type | Weight | Hits/Total |
==
| Covergroup Coverage | 1 | 100.000% |
|---------------------|--------|------------|
| Types | | 1 / 1 |
==
CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
COVERED INSTANCES: 1 / 1
FILES: 1
#
#
INSTANCE - /tb_mem_wb : work.tb_mem_wb
#
#
SUMMARY
#
==
==
| Coverage Type | Weight | Local Hits/Total | Recursive Hits/Total |

```
#
==========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |            100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
==========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
==========================================================================
========
#    |                  Covergroup                  |   Hits   |  Goal /  | Status  |
#    |                                              |          | At Least |         |
#
==========================================================================
========
#    | TYPE /tb_mem_wb/mem_wb_cg                    | 100.000% | 100.000% | Covered |
#
==========================================================================
========
#    | INSTANCE MEM_WB_COVERAGE                     | 100.000% | 100.000% | Covered |
#    |----------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_rst           | 100.000% | 100.000% | Covered |
#    |----------------------------------------------|----------|----------|---------|
#    | bin asserted                                 |        1 |        1 | Covered |
#    | bin deasserted                               |       13 |        1 | Covered |
#    |----------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_reg_write     | 100.000% | 100.000% | Covered |
#    |----------------------------------------------|----------|----------|---------|
#    | bin off                                      |        4 |        1 | Covered |
```

```
#     | bin on                                   |       10 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT MEM_WB_COVERAGE::cp_mem_to_reg    | 100.000% | 100.000% | Covered
|
#     |--------------------------------------------|----------|----------|---------|
#     | bin off                                  |        3 |        1 | Covered |
#     | bin on                                   |       11 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT MEM_WB_COVERAGE::cp_result_src    | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin alu_path                             |        2 |        1 | Covered |
#     | bin mem_path                             |        2 |        1 | Covered |
#     | bin pc4_path                             |        7 |        1 | Covered |
#     | bin default_path                         |        3 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT MEM_WB_COVERAGE::cp_rd            | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin x0                                   |        1 |        1 | Covered |
#     | bin low_reg                              |        6 |        1 | Covered |
#     | bin mid_reg                              |        1 |        1 | Covered |
#     | bin high_reg                             |        6 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | CROSS MEM_WB_COVERAGE::cross_result_regwrite | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin <alu_path,off>                       |        1 |        1 | Covered |
#     | bin <alu_path,on>                        |        1 |        1 | Covered |
#     | bin <mem_path,off>                       |        1 |        1 | Covered |
#     | bin <mem_path,on>                        |        1 |        1 | Covered |
#     | bin <pc4_path,off>                       |        1 |        1 | Covered |
#     | bin <pc4_path,on>                        |        6 |        1 | Covered |
#     | bin <default_path,off>                   |        1 |        1 | Covered |
#     | bin <default_path,on>                    |        2 |        1 | Covered |
```

```
#
================================================================================
========
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
# MODULE - work.tb_mem_wb
#    SUMMARY
#    =============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    =============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    =============================================
#    WEIGHTED AVERAGE: 100.000%
#    COVERGROUP COVERAGE
```

```
#
================================================================================
#    |                Covergroup                |  Hits   |  Goal /  | Status  |
#    |                                          |         | At Least |         |
#
================================================================================
#    | TYPE /tb_mem_wb/mem_wb_cg                | 100.000% | 100.000% | Covered |
#
================================================================================
#    | INSTANCE MEM_WB_COVERAGE                 | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_rst       | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin asserted                             |        1 |        1 | Covered |
#    | bin deasserted                           |       13 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_reg_write | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin off                                  |        4 |        1 | Covered |
#    | bin on                                   |       10 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_mem_to_reg | 100.000% | 100.000% | Covered
|
#    |------------------------------------------|----------|----------|---------|
#    | bin off                                  |        3 |        1 | Covered |
#    | bin on                                   |       11 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT MEM_WB_COVERAGE::cp_result_src | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin alu_path                             |        2 |        1 | Covered |
#    | bin mem_path                             |        2 |        1 | Covered |
#    | bin pc4_path                             |        7 |        1 | Covered |
```

```
#     | bin default_path                          |        3 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | COVERPOINT MEM_WB_COVERAGE::cp_rd           | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin x0                                    |        1 |        1 | Covered |
#     | bin low_reg                               |        6 |        1 | Covered |
#     | bin mid_reg                               |        1 |        1 | Covered |
#     | bin high_reg                              |        6 |        1 | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | CROSS MEM_WB_COVERAGE::cross_result_regwrite | 100.000% | 100.000% | Covered |
#     |--------------------------------------------|----------|----------|---------|
#     | bin <alu_path,off>                        |        1 |        1 | Covered |
#     | bin <alu_path,on>                         |        1 |        1 | Covered |
#     | bin <mem_path,off>                        |        1 |        1 | Covered |
#     | bin <mem_path,on>                         |        1 |        1 | Covered |
#     | bin <pc4_path,off>                        |        1 |        1 | Covered |
#     | bin <pc4_path,on>                         |        6 |        1 | Covered |
#     | bin <default_path,off>                    |        1 |        1 | Covered |
#     | bin <default_path,on>                     |        2 |        1 | Covered |
#
==========================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

16. Mux_wb

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_mux_wb."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

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HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 5 (71.43%) other processes in SLP

SLP: 5 (12.82%) signals in SLP and 5 (12.82%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

```
# KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.
# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.
# KERNEL: SLP simulation initialization done - time: 0.0 [s].
# KERNEL: Kernel process initialization done.
# Allocation: Simulator allocated 5300 kB (elbread=455 elab2=4835 kernel=8 sdf=0)
# KERNEL: ASDB file was created in location /home/runner/dataset.asdb
run -all
# KERNEL:
# KERNEL: TC1 : ALU Result Select
# KERNEL: ASSERT PASS : ALU PATH
# KERNEL: PASS : TC1
# KERNEL:
# KERNEL: TC2 : Memory Data Select
# KERNEL: ASSERT PASS : MEMORY PATH
# KERNEL: PASS : TC2
# KERNEL:
# KERNEL: TC3 : PC+4 Select
# KERNEL: ASSERT PASS : PC4 PATH
# KERNEL: PASS : TC3
# KERNEL:
# KERNEL: TC4 : Default Case
# KERNEL: ASSERT PASS : DEFAULT PATH
# KERNEL: PASS : TC4
# KERNEL:
# KERNEL: TC5 : Dynamic Switching
# KERNEL: PASS : TC5
# KERNEL:
# KERNEL: MUX_WB VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (187): $finish called.
# KERNEL: Time: 50 ns,  Iteration: 0,  Instance: /tb_mux_wb,  Process: @INITIAL#48_4@.
```

```
# KERNEL: stopped at time: 50 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     REPORT INFO     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property    |        Value        |
# ===============================================
# | User            | runner                  |
# | Host            | afea93787e64            |
# | Tool            | Riviera-PRO 2025.04     |
# | Report file     | /home/runner/cov.txt    |
# | Report date     | 2026-06-17 08:46        |
# | Report arguments | -verbose               |
# | Input file      | /home/runner/fcover.acdb |
# | Input file date | 2026-06-17 08:46        |
# | Number of tests | 1                       |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |            Value            |
# ==============================================
# | Test     | fcover.acdb:fcover          |
```

```
# | Status   | Ok                          |
# | Args     | asim +access+r              |
# | Simtime  | 50000 ps                    |
# | Cputime  | 0.041 s                     |
# | Seed     | 1                           |
# | Date     | 2026-06-17 08:46            |
# | User     | runner                      |
# | Host     | afea93787e64                |
# | Host os  | Linux64                     |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==============================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_mux_wb : work.tb_mux_wb
```

```
#
#
#    SUMMARY
#
============================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
============================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
============================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
============================================================================
=====
#    |                Covergroup                |   Hits   |  Goal /  | Status  |
#    |                                          |          | At Least |         |
#
============================================================================
=====
#    | TYPE /tb_mux_wb/mux_wb_cg                | 100.000% | 100.000% | Covered |
#
============================================================================
=====
#    | INSTANCE MUX_WB_COVERAGE                 | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT MUX_WB_COVERAGE::cp_result_src | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
```

```
#     | bin alu_path                          |        2 |        1 | Covered |
#     | bin mem_path                          |        2 |        1 | Covered |
#     | bin pc4_path                          |        2 |        1 | Covered |
#     | bin default_path                      |        2 |        1 | Covered |
# ==========================================================================
=====
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_mux_wb
#
#
#    SUMMARY
#    ==============================================
#    |    Coverage Type    | Weight | Hits/Total |
```

```
#     ==================================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |--------------------|--------|------------|
#     | Types               |        |      1 / 1 |
#     ==================================================
#     WEIGHTED AVERAGE: 100.000%
#
#     COVERGROUP COVERAGE
#
================================================================================
=====
#     |                Covergroup                 |   Hits   |  Goal /  | Status  |
#     |                                           |          | At Least |         |
#
================================================================================
=====
#     | TYPE /tb_mux_wb/mux_wb_cg                 | 100.000% | 100.000% | Covered |
#
================================================================================
=====
#     | INSTANCE MUX_WB_COVERAGE                  | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT MUX_WB_COVERAGE::cp_result_src | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin alu_path                              |        2 |        1 | Covered |
#     | bin mem_path                              |        2 |        1 | Covered |
#     | bin pc4_path                              |        2 |        1 | Covered |
#     | bin default_path                          |        2 |        1 | Covered |
#
================================================================================
=====
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

17. Mux

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_mux."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

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HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.0 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 3 (60.00%) other processes in SLP

SLP: 4 (10.81%) signals in SLP and 4 (10.81%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6315 kB (elbread=451 elab2=5855 kernel=8 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : PC+4 Path

KERNEL: ASSERT PASS : PC+4 PATH

KERNEL: PASS : TC1

KERNEL:

KERNEL: TC2 : Target Path

KERNEL: ASSERT PASS : TARGET PATH

KERNEL: PASS : TC2

KERNEL:

KERNEL: TC3 : Dynamic Inputs

KERNEL: ASSERT PASS : PC+4 PATH

KERNEL: PASS : TC3-1

KERNEL: ASSERT PASS : TARGET PATH

KERNEL: PASS : TC3-2

KERNEL:

KERNEL: TC4 : Zero Values

KERNEL: ASSERT PASS : PC+4 PATH

KERNEL: PASS : TC4

KERNEL:

KERNEL: TC5 : Max Values

KERNEL: ASSERT PASS : TARGET PATH

KERNEL: PASS : TC5

KERNEL:

KERNEL: MUX VERIFICATION COMPLETE

KERNEL:

```
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (195): $finish called.
# KERNEL: Time: 40 ns,  Iteration: 0,  Instance: /tb_mux,  Process: @INITIAL#80_2@.
# KERNEL: stopped at time: 40 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User              | runner                  |
# | Host              | 98a2fcb9bb3d            |
# | Tool              | Riviera-PRO 2025.04     |
# | Report file       | /home/runner/cov.txt    |
# | Report date       | 2026-06-18 14:41        |
# | Report arguments  | -verbose                |
# | Input file        | /home/runner/fcover.acdb |
# | Input file date   | 2026-06-18 14:41        |
# | Number of tests   | 1                       |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
```

```
# | Property |              Value              |
# ==================================================
# | Test     | fcover.acdb:fcover               |
# | Status   | Ok                               |
# | Args     | asim +access+r                   |
# | Simtime  | 40000 ps                         |
# | Cputime  | 0.033 s                          |
# | Seed     | 1                                |
# | Date     | 2026-06-18 14:41                 |
# | User     | runner                           |
# | Host     | 98a2fcb9bb3d                     |
# | Host os  | Linux64                          |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
```

```
#
#
# INSTANCE - /tb_mux : work.tb_mux
#
#
#     SUMMARY
#
=========================================================================
==
#     |     Coverage Type     | Weight | Local Hits/Total | Recursive Hits/Total |
#
=========================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |            100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
=========================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#     ====================================================================
#     |            Covergroup             |   Hits   |  Goal /  | Status  |
#     |                                   |          | At Least |         |
#     ====================================================================
#     | TYPE /tb_mux/mux_cg               | 100.000% | 100.000% | Covered |
#     ====================================================================
#     | INSTANCE MUX_COVERAGE             | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
#     | COVERPOINT MUX_COVERAGE::cp_pcsrc | 100.000% | 100.000% | Covered |
#     |-----------------------------------|----------|----------|---------|
```

```
#     | bin pc_plus_4_path                  |          3 |          1 | Covered |
#     | bin target_path                     |          3 |          1 | Covered |
#     ================================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++++        DESIGN UNITS        ++++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ====================================================
# |     Coverage Type     | Weight | Hits/Total |
# ====================================================
# | Covergroup Coverage |       1 |   100.000% |
# |---------------------|--------|------------|
# | Types                |         |       1 / 1 |
# ====================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_mux
#
#
#     SUMMARY
#     ====================================================
#     |     Coverage Type     | Weight | Hits/Total |
#     ====================================================
#     | Covergroup Coverage |       1 |   100.000% |
#     |---------------------|--------|------------|
```

```
#    | Types                |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#    ==========================================================================
#    |            Covergroup             |   Hits   |  Goal /  | Status  |
#    |                                   |          | At Least |         |
#    ==========================================================================
#    | TYPE /tb_mux/mux_cg               | 100.000% | 100.000% | Covered |
#    ==========================================================================
#    | INSTANCE MUX_COVERAGE             | 100.000% | 100.000% | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | COVERPOINT MUX_COVERAGE::cp_pcsrc | 100.000% | 100.000% | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | bin pc_plus_4_path                |        3 |        1 | Covered |
#    | bin target_path                   |        3 |        1 | Covered |
#    ==========================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

18. MuxSrc1

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_muxSrc1."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 5 (12.82%) signals in SLP and 5 (12.82%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5298 kB (elbread=455 elab2=4834 kernel=9 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : NORMAL PATH

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : NORMAL PATH

KERNEL:

KERNEL: TC2 : ALU FORWARD

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : ALU FORWARD

KERNEL:

KERNEL: TC3 : MEM FORWARD

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : MEM FORWARD

KERNEL:

KERNEL: TC4 : DEFAULT PATH

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : DEFAULT PATH

KERNEL:

KERNEL: TC5 : DYNAMIC SWITCHING

KERNEL: ASSERT PASS : DYNAMIC NORMAL

KERNEL: ASSERT PASS : DYNAMIC ALU

KERNEL: ASSERT PASS : DYNAMIC MEM

KERNEL: ASSERT PASS : DYNAMIC DEFAULT

KERNEL: PASS : TC5

KERNEL:

KERNEL: MUXSRC1 VERIFICATION COMPLETE

```
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (194): $finish called.
# KERNEL: Time: 50 ns,  Iteration: 0,  Instance: /tb_muxSrc1,  Process: @INITIAL#53_0@.
# KERNEL: stopped at time: 50 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User              | runner                  |
# | Host              | b2e40f7ece8e            |
# | Tool              | Riviera-PRO 2025.04     |
# | Report file       | /home/runner/cov.txt    |
# | Report date       | 2026-06-18 14:52        |
# | Report arguments  | -verbose                |
# | Input file        | /home/runner/fcover.acdb |
# | Input file date   | 2026-06-18 14:52        |
# | Number of tests   | 1                       |
# ===============================================
#
#
# TEST DETAILS
```

```
# ==============================================
# | Property |             Value             |
# ==============================================
# | Test     | fcover.acdb:fcover             |
# | Status   | Ok                             |
# | Args     | asim +access+r                 |
# | Simtime  | 50000 ps                       |
# | Cputime  | 0.036 s                        |
# | Seed     | 1                              |
# | Date     | 2026-06-18 14:52               |
# | User     | runner                         |
# | Host     | b2e40f7ece8e                   |
# | Host os  | Linux64                        |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==============================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
```

```
# FILES: 1
#
#
# INSTANCE - /tb_muxSrc1 : work.tb_muxSrc1
#
#
#    SUMMARY
#
===========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
===========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
===========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===========================================================================
=====
#    |                Covergroup                |   Hits   |  Goal /  | Status  |
#    |                                          |          | At Least |         |
#
===========================================================================
=====
#    | TYPE /tb_muxSrc1/muxSrc1_cg              | 100.000% | 100.000% | Covered |
#
===========================================================================
=====
```

```
#     | INSTANCE MUXSRC1_COVERAGE                | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT MUXSRC1_COVERAGE::cp_forward_A | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin normal_path                         |        2 |        1 | Covered |
#     | bin alu_forward                         |        2 |        1 | Covered |
#     | bin mem_forward                         |        2 |        1 | Covered |
#     | bin default_path                        |        2 |        1 | Covered |
#
==============================================================================
=====
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# +++++++++++       DESIGN UNITS      +++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_muxSrc1
#     SUMMARY
```

```
#     ================================================
#     |    Coverage Type    | Weight | Hits/Total |
#     ================================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |---------------------|--------|------------|
#     | Types               |        |      1 / 1 |
#     ================================================
#     WEIGHTED AVERAGE: 100.000%
#     COVERGROUP COVERAGE
==========================================================================
#     |              Covergroup               |   Hits   |  Goal /  | Status  |
#     |                                       |          | At Least |         |
#
==========================================================================
#     | TYPE /tb_muxSrc1/muxSrc1_cg           | 100.000% | 100.000% | Covered |
#
==========================================================================
#     | INSTANCE MUXSRC1_COVERAGE             | 100.000% | 100.000% | Covered |
#     |---------------------------------------|----------|----------|---------|
#     | COVERPOINT MUXSRC1_COVERAGE::cp_forward_A | 100.000% | 100.000% | Covered |
#     |---------------------------------------|----------|----------|---------|
#     | bin normal_path                       |        2 |        1 | Covered |
#     | bin alu_forward                       |        2 |        1 | Covered |
#     | bin mem_forward                       |        2 |        1 | Covered |
#     | bin default_path                      |        2 |        1 | Covered |
#
==========================================================================
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

19.MuxSrc2

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_muxSrc2."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 5 (12.82%) signals in SLP and 5 (12.82%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5298 kB (elbread=455 elab2=4834 kernel=9 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : NORMAL PATH

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : NORMAL PATH

KERNEL:

KERNEL: TC2 : ALU FORWARD

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : ALU FORWARD

KERNEL:

KERNEL: TC3 : MEM FORWARD

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : MEM FORWARD

KERNEL:

KERNEL: TC4 : DEFAULT PATH

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : DEFAULT PATH

KERNEL:

KERNEL: TC5 : DYNAMIC SWITCHING

KERNEL: ASSERT PASS : DYNAMIC NORMAL

KERNEL: ASSERT PASS : DYNAMIC ALU

KERNEL: ASSERT PASS : DYNAMIC MEM

KERNEL: ASSERT PASS : DYNAMIC DEFAULT

KERNEL: PASS : TC5

KERNEL:

KERNEL: MUXSRC2 VERIFICATION COMPLETE

```
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (193): $finish called.
# KERNEL: Time: 50 ns,  Iteration: 0,  Instance: /tb_muxSrc2,  Process: @INITIAL#52_0@.
# KERNEL: stopped at time: 50 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User              | runner                   |
# | Host              | 86d5ff9279e0             |
# | Tool              | Riviera-PRO 2025.04      |
# | Report file       | /home/runner/cov.txt     |
# | Report date       | 2026-06-18 15:01         |
# | Report arguments  | -verbose                 |
# | Input file        | /home/runner/fcover.acdb |
# | Input file date   | 2026-06-18 15:01         |
# | Number of tests   | 1                        |
# ===============================================
#
#
# TEST DETAILS
```

```
# ================================================
# | Property |              Value              |
# ================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Ok                              |
# | Args     | asim +access+r                  |
# | Simtime  | 50000 ps                        |
# | Cputime  | 0.041 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-18 15:01                |
# | User     | runner                          |
# | Host     | 86d5ff9279e0                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
```

```
# FILES: 1
#
#
# INSTANCE - /tb_muxSrc2 : work.tb_muxSrc2
#
#
#    SUMMARY
#
===========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
===========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
===========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
===========================================================================
=====
#    |                Covergroup                |   Hits   |  Goal /  | Status  |
#    |                                          |          | At Least |         |
#
===========================================================================
=====
#    | TYPE /tb_muxSrc2/muxSrc2_cg              | 100.000% | 100.000% | Covered |
#
===========================================================================
=====
```

```
#     | INSTANCE MUXSRC2_COVERAGE                 | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT MUXSRC2_COVERAGE::cp_forward_B | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin normal_path                         |        2 |        1 | Covered |
#     | bin alu_forward                         |        2 |        1 | Covered |
#     | bin mem_forward                         |        2 |        1 | Covered |
#     | bin default_path                        |        2 |        1 | Covered |
#
================================================================================
=====
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==================================================
# |    Coverage Type    | Weight | Hits/Total |
# ==================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
# MODULE - work.tb_muxSrc2
#     SUMMARY
#     ==================================================
#     |    Coverage Type    | Weight | Hits/Total |
```

```
#     ================================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |--------------------|--------|------------|
#     | Types              |        |      1 / 1 |
#     ================================================
#     WEIGHTED AVERAGE: 100.000%
#     COVERGROUP COVERAGE
#
==========================================================================
=====
#     |              Covergroup              |   Hits   |  Goal /  | Status  |
#     |                                      |          | At Least |         |
#
==========================================================================
=====
#     | TYPE /tb_muxSrc2/muxSrc2_cg              | 100.000% | 100.000% | Covered |
#
==========================================================================
=====
#     | INSTANCE MUXSRC2_COVERAGE                | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT MUXSRC2_COVERAGE::cp_forward_B | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin normal_path                          |        2 |        1 | Covered |
#     | bin alu_forward                          |        2 |        1 | Covered |
#     | bin mem_forward                          |        2 |        1 | Covered |
#     | bin default_path                         |        2 |        1 | Covered |
#
==========================================================================
=====
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

20. MuxSrcImm

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_muxSrcImm."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 4 (10.81%) signals in SLP and 4 (10.81%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5299 kB (elbread=459 elab2=4831 kernel=8 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : Register Path

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : REGISTER PATH

KERNEL:

KERNEL: TC2 : Immediate Path

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : IMMEDIATE PATH

KERNEL:

KERNEL: TC3 : Dynamic Switching

KERNEL: ASSERT PASS : DYNAMIC REGISTER

KERNEL: ASSERT PASS : DYNAMIC IMMEDIATE

KERNEL: PASS : TC3

KERNEL:

KERNEL: TC4 : Zero Values

KERNEL: PASS : TC4

KERNEL:

KERNEL: TC5 : Large Values

KERNEL: PASS : TC5

KERNEL:

KERNEL: MUXSRCIMM VERIFICATION COMPLETE

KERNEL:

KERNEL: Coverage = 100.00%

RUNTIME: Info: RUNTIME_0068 testbench.sv (170): $finish called.

KERNEL: Time: 40 ns, Iteration: 0, Instance: /tb_muxSrcImm, Process: @INITIAL#48_0@.

```
# KERNEL: stopped at time: 40 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     REPORT INFO     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property    |         Value          |
# ===============================================
# | User           | runner                 |
# | Host           | f2f39d18c4dd           |
# | Tool           | Riviera-PRO 2025.04    |
# | Report file    | /home/runner/cov.txt   |
# | Report date    | 2026-06-18 15:07       |
# | Report arguments | -verbose             |
# | Input file     | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 15:07     |
# | Number of tests  | 1                    |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |            Value            |
# ==============================================
# | Test     | fcover.acdb:fcover           |
```

```
# | Status   | Ok                       |
# | Args     | asim +access+r                 |
# | Simtime  | 40000 ps                      |
# | Cputime  | 0.027 s                     |
# | Seed     | 1                          |
# | Date     | 2026-06-18 15:07               |
# | User     | runner                      |
# | Host     | f2f39d18c4dd                  |
# | Host os  | Linux64                      |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ===================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==================================================
# |    Coverage Type    | Weight | Hits/Total |
# ==================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_muxSrcImm : work.tb_muxSrcImm
```

```
#
#
#    SUMMARY
#
==========================================================================
==
#    |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
==========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
==========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
==========================================================================
=====
#    |                Covergroup                 |   Hits   |  Goal /  | Status  |
#    |                                           |          | At Least |         |
#
==========================================================================
=====
#    | TYPE /tb_muxSrcImm/muxSrcImm_cg           | 100.000% | 100.000% | Covered |
#
==========================================================================
=====
#    | INSTANCE MUXSRCIMM_COVERAGE               | 100.000% | 100.000% | Covered |
#    |-------------------------------------------|----------|----------|---------|
#    | COVERPOINT MUXSRCIMM_COVERAGE::cp_alu_src | 100.000% | 100.000% | Covered |
#    |-------------------------------------------|----------|----------|---------|
```

```
#     | bin reg_path                          |        3 |        1 | Covered |
#     | bin imm_path                          |        3 |        1 | Covered |
#
=========================================================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_muxSrcImm
#
#
#    SUMMARY
#    ==============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ==============================================
#    | Covergroup Coverage |      1 |   100.000% |
```

```
#     |--------------------|--------|------------|
#     | Types              |        |      1 / 1 |
#     ================================================
#     WEIGHTED AVERAGE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
=====================================================================================
#     |              Covergroup              |   Hits   |  Goal /  | Status  |
#     |                                      |          | At Least |         |
#
=====================================================================================
#     | TYPE /tb_muxSrcImm/muxSrcImm_cg          | 100.000% | 100.000% | Covered |
#
=====================================================================================
#     | INSTANCE MUXSRCIMM_COVERAGE             | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT MUXSRCIMM_COVERAGE::cp_alu_src | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin reg_path                            |        3 |        1 | Covered |
#     | bin imm_path                            |        3 |        1 | Covered |
#
=====================================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

21.PC

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_pc."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.0 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 2 (28.57%) other processes in SLP

SLP: 5 (8.62%) signals in SLP and 5 (8.62%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6338 kB (elbread=455 elab2=5873 kernel=10 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : RESET

KERNEL: SVA PASS : UPDATE

KERNEL: SVA PASS : STALL

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : RESET

KERNEL:

KERNEL: TC2 : NORMAL UPDATE

KERNEL: SVA PASS : RESET

KERNEL: SVA PASS : RESET

KERNEL: SVA PASS : STALL

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : UPDATE

KERNEL:

KERNEL: TC3 : STALL

KERNEL: SVA PASS : RESET

KERNEL: SVA PASS : UPDATE

KERNEL: SVA PASS : UPDATE

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : STALL

KERNEL:

KERNEL: TC4 : SECOND UPDATE

KERNEL: SVA PASS : RESET

KERNEL: SVA PASS : STALL

KERNEL: SVA PASS : STALL

```
# KERNEL: PASS : TC4
# KERNEL: ASSERT PASS : SECOND UPDATE
# KERNEL:
# KERNEL: TC5 : LARGE VALUE
# KERNEL: SVA PASS : RESET
# KERNEL: SVA PASS : UPDATE
# KERNEL: SVA PASS : STALL
# KERNEL: PASS : TC5
# KERNEL: ASSERT PASS : LARGE VALUE
# KERNEL:
# KERNEL: TC6 : RESET AGAIN
# KERNEL: SVA PASS : UPDATE
# KERNEL: SVA PASS : UPDATE
# KERNEL: SVA PASS : STALL
# KERNEL: PASS : TC6
# KERNEL: ASSERT PASS : RESET AGAIN
# KERNEL:
# KERNEL: PC VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# KERNEL: SVA PASS : RESET
# KERNEL: SVA PASS : UPDATE
# KERNEL: SVA PASS : STALL
# RUNTIME: Info: RUNTIME_0068 testbench.sv (232): $finish called.
# KERNEL: Time: 72 ns,  Iteration: 0,  Instance: /tb_pc,  Process: @INITIAL#92_1@.
# KERNEL: stopped at time: 72 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
```

```
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# =================================================
# |    Property    |         Value          |
# =================================================
# | User             | runner                  |
# | Host             | c2b9e90398be            |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-18 15:34         |
# | Report arguments | -verbose                |
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 15:34         |
# | Number of tests  | 1                       |
# =================================================
#
#
# TEST DETAILS
# ==============================================
# | Property |             Value             |
# ==============================================
# | Test     | fcover.acdb:fcover            |
# | Status   | Ok                           |
# | Args     | asim +access+r                |
# | Simtime  | 72000 ps                      |
# | Cputime  | 0.045 s                       |
# | Seed     | 1                            |
# | Date     | 2026-06-18 15:34              |
# | User     | runner                       |
```

```
# | Host     | c2b9e90398be                 |
# | Host os  | Linux64                      |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# =================================================
#
#
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_pc : work.tb_pc
#
#
#    SUMMARY
#
============================================================================
==
#     |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
```

```
#
========================================================================
==
#    | Covergroup Coverage |      1 |         100.000% |             100.000% |
#    |---------------------|--------|------------------|----------------------|
#    | Types               |        |            1 / 1 |                1 / 1 |
#
========================================================================
==
#    WEIGHTED AVERAGE LOCAL: 100.000%
#    WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#    COVERGROUP COVERAGE
#    ======================================================================
#    |            Covergroup             |   Hits   |  Goal /  | Status  |
#    |                                   |          | At Least |         |
#    ======================================================================
#    | TYPE /tb_pc/pc_cg                 | 100.000% | 100.000% | Covered |
#    ======================================================================
#    | INSTANCE PC_COVERAGE              | 100.000% | 100.000% | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | COVERPOINT PC_COVERAGE::cp_rst    | 100.000% | 100.000% | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | bin reset_on                      |        3 |        1 | Covered |
#    | bin reset_off                     |        4 |        1 | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | COVERPOINT PC_COVERAGE::cp_write  | 100.000% | 100.000% | Covered |
#    |-----------------------------------|----------|----------|---------|
#    | bin write_enable                  |        5 |        1 | Covered |
#    | bin write_disable                 |        2 |        1 | Covered |
#    ======================================================================
#
#
```

```
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN UNITS     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_pc
#
#
#    SUMMARY
#    ==============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ==============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ==============================================
#    WEIGHTED AVERAGE: 100.000%
#
#
```

```
#    COVERGROUP COVERAGE
#    ================================================================
#    |          Covergroup           |   Hits   |  Goal /  | Status  |
#    |                               |          | At Least |         |
#    ================================================================
#    | TYPE /tb_pc/pc_cg             | 100.000% | 100.000% | Covered |
#    ================================================================
#    | INSTANCE PC_COVERAGE          | 100.000% | 100.000% | Covered |
#    |---------------------------------|----------|----------|---------|
#    | COVERPOINT PC_COVERAGE::cp_rst   | 100.000% | 100.000% | Covered |
#    |---------------------------------|----------|----------|---------|
#    | bin reset_on                  |        3 |        1 | Covered |
#    | bin reset_off                 |        4 |        1 | Covered |
#    |---------------------------------|----------|----------|---------|
#    | COVERPOINT PC_COVERAGE::cp_write | 100.000% | 100.000% | Covered |
#    |---------------------------------|----------|----------|---------|
#    | bin write_enable              |        5 |        1 | Covered |
#    | bin write_disable             |        2 |        1 | Covered |
#    ================================================================
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

22. Pc_offset

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "Unit top modules: tb_pc_offset."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 1 (33.33%) other processes in SLP

SLP: 3 (6.25%) signals in SLP and 3 (6.25%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 5308 kB (elbread=459 elab2=4839 kernel=9 sdf=0)

ACDB: Warning: ACDB_0022 testbench.sv (44): An interval '[-1000:-1]' is completely out of coverpoint 'cp_imm' range and will be ignored. HDL instance: "/tb_pc_offset". Covergroup type: "pc_offset_cg", covergroup instance: "PC_OFFSET_COVERAGE", coverpoint: "cp_imm".

ACDB: Warning: ACDB_0030 testbench.sv (44): Bin(s) 'negative_imm' are empty and will be excluded from the coverage report. HDL instance: "/tb_pc_offset". Covergroup type: "pc_offset_cg", covergroup instance: "PC_OFFSET_COVERAGE", coverpoint: "cp_imm".

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : ZERO + ZERO

KERNEL: PASS : TC1

KERNEL: ASSERT PASS : ZERO

KERNEL:

KERNEL: TC2 : POSITIVE OFFSET

KERNEL: PASS : TC2

KERNEL: ASSERT PASS : POSITIVE OFFSET

KERNEL:

KERNEL: TC3 : NEGATIVE OFFSET

KERNEL: PASS : TC3

KERNEL: ASSERT PASS : NEGATIVE OFFSET

KERNEL:

KERNEL: TC4 : LARGE VALUES

KERNEL: PASS : TC4

KERNEL: ASSERT PASS : LARGE VALUES

KERNEL:

KERNEL: TC5 : OVERFLOW

KERNEL: PASS : TC5

KERNEL: ASSERT PASS : OVERFLOW

```
# KERNEL:
# KERNEL: TC6 : DYNAMIC SWITCHING
# KERNEL: ASSERT PASS : DYNAMIC-1
# KERNEL: ASSERT PASS : DYNAMIC-2
# KERNEL: PASS : TC6
# KERNEL:
# KERNEL: PC_OFFSET VERIFICATION COMPLETE
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (207): $finish called.
# KERNEL: Time: 45 ns,  Iteration: 0,  Instance: /tb_pc_offset,  Process: @INITIAL#59_0@.
# KERNEL: stopped at time: 45 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value          |
# ===============================================
# | User             | runner                  |
# | Host             | 0a92143f2de4            |
# | Tool             | Riviera-PRO 2025.04     |
# | Report file      | /home/runner/cov.txt    |
# | Report date      | 2026-06-18 15:23        |
# | Report arguments | -verbose                |
```

```
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-18 15:23         |
# | Number of tests  | 1                        |
# ==================================================
#
#
# TEST DETAILS
# =================================================
# | Property |              Value              |
# =================================================
# | Test     | fcover.acdb:fcover              |
# | Status   | Warning                         |
# | Args     | asim +access+r                  |
# | Simtime  | 45000 ps                        |
# | Cputime  | 0.021 s                         |
# | Seed     | 1                               |
# | Date     | 2026-06-18 15:23                |
# | User     | runner                          |
# | Host     | 0a92143f2de4                    |
# | Host os  | Linux64                         |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# =================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
```

==

| Covergroup Coverage | 1 | 100.000% |

|---------------------|--------|------------|

| Types | | 1 / 1 |

==

CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%

COVERED INSTANCES: 1 / 1

FILES: 1

#

#

INSTANCE - /tb_pc_offset : work.tb_pc_offset

#

#

SUMMARY

==

| Coverage Type | Weight | Local Hits/Total | Recursive Hits/Total |

==

| Covergroup Coverage | 1 | 100.000% | 100.000% |

|---------------------|--------|------------------|----------------------|

| Types | | 1 / 1 | 1 / 1 |

==

WEIGHTED AVERAGE LOCAL: 100.000%

WEIGHTED AVERAGE RECURSIVE: 100.000%

#

#

COVERGROUP COVERAGE

===

```
#    |              Covergroup              |   Hits   |  Goal /  | Status  |
#    |                                  |          | At Least |         |
#
=
=
#    | TYPE /tb_pc_offset/pc_offset_cg       | 100.000% | 100.000% | Covered |
#
=
=
#    | INSTANCE PC_OFFSET_COVERAGE           | 100.000% | 100.000% | Covered |
#    |--------------------------------------|----------|----------|---------|
#    | COVERPOINT PC_OFFSET_COVERAGE::cp_pc  | 100.000% | 100.000% | Covered |
#    |--------------------------------------|----------|----------|---------|
#    | bin zero_pc                          |        1 |        1 | Covered |
#    | bin small_pc                         |        2 |        1 | Covered |
#    | bin large_pc                         |        4 |        1 | Covered |
#    |--------------------------------------|----------|----------|---------|
#    | COVERPOINT PC_OFFSET_COVERAGE::cp_imm | 100.000% | 100.000% | Covered |
#    |--------------------------------------|----------|----------|---------|
#    | bin zero_imm                         |        1 |        1 | Covered |
#    | bin positive_imm                     |        3 |        1 | Covered |
#    | bin large_imm                        |        3 |        1 | Covered |
#
=
=
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ============================================
```

```
# |    Coverage Type    | Weight | Hits/Total |
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_pc_offset
#
#
#    SUMMARY
#    ================================================
#    |    Coverage Type    | Weight | Hits/Total |
#    ================================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
#    ================================================
#    WEIGHTED AVERAGE: 100.000%
#
#
#    COVERGROUP COVERAGE
#
================================================================================
=
#    |              Covergroup              |  Hits   |  Goal /  | Status  |
#    |                                      |         | At Least |         |
#
================================================================================
=
```

```
#     | TYPE /tb_pc_offset/pc_offset_cg       | 100.000% | 100.000% | Covered |
#
=======================================================================
=
#     | INSTANCE PC_OFFSET_COVERAGE          | 100.000% | 100.000% | Covered |
#     |--------------------------------------|----------|----------|---------|
#     | COVERPOINT PC_OFFSET_COVERAGE::cp_pc  | 100.000% | 100.000% | Covered |
#     |--------------------------------------|----------|----------|---------|
#     | bin zero_pc                          |        1 |        1 | Covered |
#     | bin small_pc                         |        2 |        1 | Covered |
#     | bin large_pc                         |        4 |        1 | Covered |
#     |--------------------------------------|----------|----------|---------|
#     | COVERPOINT PC_OFFSET_COVERAGE::cp_imm | 100.000% | 100.000% | Covered |
#     |--------------------------------------|----------|----------|---------|
#     | bin zero_imm                         |        1 |        1 | Covered |
#     | bin positive_imm                     |        3 |        1 | Covered |
#     | bin large_imm                        |        3 |        1 | Covered |
#
=======================================================================
=
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

23.Regfile

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE "ASRT_7002: Due to optimizations some assertions from source code will be evaluated in non LRM complaint mode. Use switch -ao_off to turn off this optimization."

MESSAGE "Unit top modules: tb_reg_file."

SUCCESS "Compile success 0 Errors 0 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

SLP: 0 primitives and 4 (40.00%) other processes in SLP

SLP: 10 (6.85%) signals in SLP and 10 (6.85%) interface signals

ELAB2: Elaboration final pass complete - time: 0.1 [s].

KERNEL: SLP loading done - time: 0.0 [s].

KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

KERNEL: SLP simulation initialization done - time: 0.0 [s].

KERNEL: Kernel process initialization done.

Allocation: Simulator allocated 6401 kB (elbread=459 elab2=5928 kernel=13 sdf=0)

KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

KERNEL:

KERNEL: TC1 : Reset Verification

KERNEL: PASS : x0 Protection

KERNEL: PASS : TC1

KERNEL:

KERNEL: TC2 : Single Register Write

KERNEL: PASS : x0 remains zero

KERNEL: PASS : x0 Protection

KERNEL: PASS : Reset Clear

KERNEL: PASS : Reset Clear

KERNEL: PASS : Register Write

KERNEL: PASS : Register Write

KERNEL: PASS : x0 Protection

KERNEL: PASS : Reset Clear

KERNEL: PASS : TC2

KERNEL:

KERNEL: TC3 : Dual Read Verification

KERNEL: PASS : x0 Protection

KERNEL: PASS : Reset Clear

KERNEL: PASS : TC3

KERNEL:

KERNEL: TC4 : Write Disable Verification

KERNEL: PASS : Register Write

KERNEL: PASS : Register Write

```
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : TC4
# KERNEL:
# KERNEL: TC5 : x0 Protection
# KERNEL: PASS : Register Write
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : TC5
# KERNEL:
# KERNEL: TC6 : Multiple Register Writes
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : x1
# KERNEL: PASS : Register Write
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : x2
# KERNEL: PASS : x3
# KERNEL:
# KERNEL: TC7 : Register Update
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Update Step 1
# KERNEL: PASS : Register Write
```

```
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Update Step 2
# KERNEL:
# KERNEL: TC8 : Coverage Closure
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : x0 Protection
# KERNEL:
# KERNEL: TC9 : Mid Register Coverage
# KERNEL: PASS : x0 remains zero
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : rs1 mid_reg
# KERNEL: PASS : rs2 mid_reg
# KERNEL:
# KERNEL: REG_FILE VERIFICATION COMPLETE
```

```
# KERNEL: PASS : Register Write
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL: PASS : Register Write
# KERNEL: PASS : x0 Protection
# KERNEL: PASS : Reset Clear
# KERNEL:
# KERNEL: Coverage = 100.00%
# RUNTIME: Info: RUNTIME_0068 testbench.sv (486): $finish called.
# KERNEL: Time: 214 ns,  Iteration: 0,  Instance: /tb_reg_file,  Process: @INITIAL#183_1@.
# KERNEL: stopped at time: 214 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
exec cat cov.txt
# ++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |     Property     |          Value           |
# ===============================================
# | User             | runner                   |
# | Host             | 1ee9b689b026             |
# | Tool             | Riviera-PRO 2025.04      |
# | Report file      | /home/runner/cov.txt     |
# | Report date      | 2026-06-17 08:03         |
# | Report arguments | -verbose                 |
```

```
# | Input file       | /home/runner/fcover.acdb |
# | Input file date  | 2026-06-17 08:03         |
# | Number of tests  | 1                        |
# ==================================================
#
#
# TEST DETAILS
# ==================================================
# | Property |              Value               |
# ==================================================
# | Test     | fcover.acdb:fcover               |
# | Status   | Ok                               |
# | Args     | asim +access+r                   |
# | Simtime  | 214000 ps                        |
# | Cputime  | 0.107 s                          |
# | Seed     | 1                                |
# | Date     | 2026-06-17 08:03                 |
# | User     | runner                           |
# | Host     | 1ee9b689b026                     |
# | Host os  | Linux64                          |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ==================================================
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY     ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ================================================
# |    Coverage Type    | Weight | Hits/Total |
```

```
# ================================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ================================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_reg_file : work.tb_reg_file
#
#
#     SUMMARY
#
================================================================================
==
#     |    Coverage Type    | Weight | Local Hits/Total | Recursive Hits/Total |
#
================================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
================================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
================================================================================
=====
```

```
#     |                 Covergroup                 |   Hits    |  Goal /  | Status  |
#     |                                            |           | At Least |         |
#
=============================================================================
=====
#     | TYPE /tb_reg_file/regfile_cg               | 100.000% | 100.000% | Covered |
#
=============================================================================
=====
#     | INSTANCE REGFILE_COVERAGE                  | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_rst        | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin asserted                               |        2 |        1 | Covered |
#     | bin deasserted                             |       19 |        1 | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_reg_write | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin off                                    |        8 |        1 | Covered |
#     | bin on                                     |       13 |        1 | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_rd         | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin x0                                     |        2 |        1 | Covered |
#     | bin low_reg                                |       10 |        1 | Covered |
#     | bin mid_reg                                |        5 |        1 | Covered |
#     | bin high_reg                               |        4 |        1 | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_rs1        | 100.000% | 100.000% | Covered |
#     |-----------------------------------------|----------|----------|---------|
#     | bin x0                                     |        4 |        1 | Covered |
#     | bin low_reg                                |       10 |        1 | Covered |
#     | bin mid_reg                                |        2 |        1 | Covered |
#     | bin high_reg                               |        5 |        1 | Covered |
```

```
#    |----------------------------------------|----------|----------|---------|
#    | COVERPOINT REGFILE_COVERAGE::cp_rs2     | 100.000% | 100.000% | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | bin x0                                  |        4 |        1 | Covered |
#    | bin low_reg                             |       11 |        1 | Covered |
#    | bin mid_reg                             |        2 |        1 | Covered |
#    | bin high_reg                            |        4 |        1 | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | COVERPOINT REGFILE_COVERAGE::cp_data    | 100.000% | 100.000% | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | bin data_low                            |        7 |        1 | Covered |
#    | bin data_mid                            |       10 |        1 | Covered |
#    | bin data_high                           |        4 |        1 | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | CROSS REGFILE_COVERAGE::cross_write_rd  | 100.000% | 100.000% | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | bin <off,x0>                            |        1 |        1 | Covered |
#    | bin <off,low_reg>                       |        3 |        1 | Covered |
#    | bin <off,mid_reg>                       |        3 |        1 | Covered |
#    | bin <off,high_reg>                      |        1 |        1 | Covered |
#    | bin <on,x0>                             |        1 |        1 | Covered |
#    | bin <on,low_reg>                        |        7 |        1 | Covered |
#    | bin <on,mid_reg>                        |        2 |        1 | Covered |
#    | bin <on,high_reg>                       |        3 |        1 | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | CROSS REGFILE_COVERAGE::cross_rst_write | 100.000% | 100.000% | Covered |
#    |----------------------------------------|----------|----------|---------|
#    | bin <asserted,off>                      |        1 |        1 | Covered |
#    | bin <asserted,on>                       |        1 |        1 | Covered |
#    | bin <deasserted,off>                    |        7 |        1 | Covered |
#    | bin <deasserted,on>                     |       12 |        1 | Covered |
```

```
#
=====================================================================================
=====
#
#
# ++++++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# ++++++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_reg_file
#
#
#    SUMMARY
#    =============================================
#    |    Coverage Type    | Weight | Hits/Total |
#    =============================================
#    | Covergroup Coverage |      1 |   100.000% |
#    |---------------------|--------|------------|
#    | Types               |        |      1 / 1 |
```

```
#     ==================================================
#     WEIGHTED AVERAGE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
================================================================================
=====
#     |              Covergroup              |   Hits   |  Goal /  | Status  |
#     |                                      |          | At Least |         |
#
================================================================================
=====
#     | TYPE /tb_reg_file/regfile_cg             | 100.000% | 100.000% | Covered |
#
================================================================================
=====
#     | INSTANCE REGFILE_COVERAGE                | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_rst       | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin asserted                             |        2 |        1 | Covered |
#     | bin deasserted                           |       19 |        1 | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_reg_write | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin off                                  |        8 |        1 | Covered |
#     | bin on                                   |       13 |        1 | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT REGFILE_COVERAGE::cp_rd        | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin x0                                   |        2 |        1 | Covered |
#     | bin low_reg                              |       10 |        1 | Covered |
#     | bin mid_reg                              |        5 |        1 | Covered |
```

```
#    | bin high_reg                        |        4 |        1 | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | COVERPOINT REGFILE_COVERAGE::cp_rs1       | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | bin x0                              |        4 |        1 | Covered |
#    | bin low_reg                         |       10 |        1 | Covered |
#    | bin mid_reg                         |        2 |        1 | Covered |
#    | bin high_reg                        |        5 |        1 | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | COVERPOINT REGFILE_COVERAGE::cp_rs2       | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | bin x0                              |        4 |        1 | Covered |
#    | bin low_reg                         |       11 |        1 | Covered |
#    | bin mid_reg                         |        2 |        1 | Covered |
#    | bin high_reg                        |        4 |        1 | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | COVERPOINT REGFILE_COVERAGE::cp_data      | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | bin data_low                        |        7 |        1 | Covered |
#    | bin data_mid                        |       10 |        1 | Covered |
#    | bin data_high                       |        4 |        1 | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | CROSS REGFILE_COVERAGE::cross_write_rd    | 100.000% | 100.000% | Covered |
#    |-----------------------------------------|----------|----------|---------|
#    | bin <off,x0>                        |        1 |        1 | Covered |
#    | bin <off,low_reg>                   |        3 |        1 | Covered |
#    | bin <off,mid_reg>                   |        3 |        1 | Covered |
#    | bin <off,high_reg>                  |        1 |        1 | Covered |
#    | bin <on,x0>                         |        1 |        1 | Covered |
#    | bin <on,low_reg>                    |        7 |        1 | Covered |
#    | bin <on,mid_reg>                    |        2 |        1 | Covered |
#    | bin <on,high_reg>                   |        3 |        1 | Covered |
```

```
#     |------------------------------------------|----------|----------|---------|
#     | CROSS REGFILE_COVERAGE::cross_rst_write   | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin <asserted,off>                        |        1 |        1 | Covered |
#     | bin <asserted,on>                         |        1 |        1 | Covered |
#     | bin <deasserted,off>                      |        7 |        1 | Covered |
#     | bin <deasserted,on>                       |       12 |        1 | Covered |
#
=============================================================================
=====
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```

24. Risc_top

VSIMSA: Configuration file changed: `/home/runner/library.cfg'

ALIB: Library "work" attached.

work = ./work/work.lib

MESSAGE_SP VCP2876 "Implicit net declaration, symbol ID_EX_write has not been declared in module risc_top." "design.sv" 1672 12

WARNING VCP2597 "Some unconnected ports remain at instance: ""id_ex1"". Module ""ID_EX"" has unconnected port(s) : ""id_ex_write""." "design.sv" 1461 1

MESSAGE "ASRT_7002: Due to optimizations some assertions from source code will be evaluated in non LRM complaint mode. Use switch -ao_off to turn off this optimization."

MESSAGE "Unit top modules: tb_risc_top."

SUCCESS "Compile success 0 Errors 1 Warnings Analysis time: 0[s]."

done

Aldec, Inc. Riviera-PRO version 2025.04.139.9738 built for Linux64 on May 30, 2025.

HDL, SystemC, and Assertions simulator, debugger, and design environment.

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vsim +access+r

ELBREAD: Elaboration process.

ELBREAD: Elaboration time 0.0 [s].

KERNEL: Main thread initiated.

KERNEL: Kernel process initialization phase.

ELAB2: Elaboration final pass...

KERNEL: PLI/VHPI kernel's engine initialization done.

PLI: Loading library '/usr/share/Riviera-PRO/bin/libsystf.so'

ELAB2: Create instances ...

KERNEL: Time resolution set to 1ps.

ELAB2: Create instances complete.

SLP: Started

SLP: Elaboration phase ...

SLP: Elaboration phase ... done : 0.0 [s]

SLP: Generation phase ...

SLP: Generation phase ... done : 0.1 [s]

SLP: Finished : 0.1 [s]

```
# SLP: 0 primitives and 36 (76.60%) other processes in SLP

# SLP: 279 (65.65%) signals in SLP and 18 (4.24%) interface signals

# ELAB2: Elaboration final pass complete - time: 0.2 [s].

# KERNEL: SLP loading done - time: 0.0 [s].

# KERNEL: Warning: You are using the Riviera-PRO EDU Edition. The performance of simulation is reduced.

# KERNEL: Warning: Contact Aldec for available upgrade options - sales@aldec.com.

# KERNEL: SLP simulation initialization done - time: 0.0 [s].

# KERNEL: Kernel process initialization done.

# Allocation: Simulator allocated 6527 kB (elbread=459 elab2=6054 kernel=13 sdf=0)

# KERNEL: ASDB file was created in location /home/runner/dataset.asdb

run -all

# KERNEL:

# KERNEL: ================================

# KERNEL: TC1 : RESET

# KERNEL: ================================

# KERNEL: ASSERT PASS : STALL

# KERNEL: ASSERT PASS : BRANCH FLUSH

# KERNEL: ASSERT PASS : EX FORWARD A

# KERNEL: ASSERT PASS : MEM FORWARD A

# KERNEL: ASSERT PASS : MEM READ

# KERNEL: ASSERT PASS : RESET PC

# KERNEL: ASSERT PASS : RESET PC

# KERNEL: ASSERT PASS : PC UPDATE

# KERNEL: ASSERT PASS : STALL

# KERNEL: ASSERT PASS : BRANCH FLUSH

# KERNEL: ASSERT PASS : WRITEBACK

# KERNEL: ASSERT PASS : RESET PC

# KERNEL: ASSERT PASS : RESET PC

# KERNEL: ASSERT PASS : PC UPDATE

# KERNEL: ASSERT PASS : STALL

# KERNEL: ASSERT PASS : WRITEBACK

# KERNEL: PC = 4
```

```
# KERNEL:
# KERNEL: ================================
# KERNEL: TC2 : INSTRUCTION FETCH
# KERNEL: ================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: TIME=55000 PC=00000008 INSTR=002081b3
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: TIME=65000 PC=0000000c INSTR=00118233
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: TIME=75000 PC=00000010 INSTR=002202b3
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : MEM FORWARD A
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: TIME=85000 PC=00000014 INSTR=00502023
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : EX FORWARD A
# KERNEL: ASSERT PASS : MEM FORWARD A
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: TIME=95000 PC=00000018 INSTR=00002303
# KERNEL:
```

KERNEL: ================================

KERNEL: TC3 : PIPELINE FLOW

KERNEL: ================================

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=0000001c IF=00002383 ID=00002303 EX_RD=0 MEM_RD=5 WB_RD=4

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : EX FORWARD A

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000020 IF=00138433 ID=00002383 EX_RD=6 MEM_RD=0 WB_RD=5

KERNEL: WRITE: MEM[0] = 20

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000024 IF=00208463 ID=00138433 EX_RD=7 MEM_RD=6 WB_RD=0

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : MEM READ

KERNEL: PC=00000024 IF=00208463 ID=00138433 EX_RD=7 MEM_RD=7 WB_RD=6

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000028 IF=00108463 ID=00208463 EX_RD=8 MEM_RD=7 WB_RD=7

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : MEM FORWARD A

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : MEM READ

KERNEL: PC=0000002c IF=0080006f ID=00108463 EX_RD=8 MEM_RD=8 WB_RD=7

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : MEM FORWARD A

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000030 IF=00100093 ID=0080006f EX_RD=8 MEM_RD=8 WB_RD=8

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : BRANCH FLUSH

KERNEL: ASSERT PASS : BRANCH FLUSH

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000030 IF=00100093 ID=00000000 EX_RD=0 MEM_RD=8 WB_RD=8

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : BRANCH FLUSH

KERNEL: ASSERT PASS : BRANCH FLUSH

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000034 IF=00200113 ID=00100093 EX_RD=0 MEM_RD=0 WB_RD=8

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: PC=00000038 IF=002081b3 ID=00200113 EX_RD=1 MEM_RD=0 WB_RD=0

KERNEL:

KERNEL: ================================

KERNEL: TC4 : WRITEBACK

KERNEL: ================================

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: WB -> rd=1 data=00000001

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : MEM FORWARD A

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: WB -> rd=2 data=00000002

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : MEM FORWARD A

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: WB -> rd=3 data=00000003

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: WB -> rd=0 data=00000000

KERNEL: ASSERT PASS : RESET PC

```
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=4 data=0000000d
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: WB -> rd=0 data=00000000
# KERNEL:
# KERNEL: ================================
# KERNEL: TC5 : MEMORY
# KERNEL: ================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL:
# KERNEL: ================================
# KERNEL: TC6 : HAZARD CHECK
# KERNEL: ================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL:
# KERNEL: ================================
# KERNEL: TC7 : FORWARDING CHECK
# KERNEL: ================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

KERNEL: ASSERT PASS : WRITEBACK

KERNEL: ASSERT PASS : RESET PC

KERNEL: ASSERT PASS : PC UPDATE

KERNEL: ASSERT PASS : STALL

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL:
# KERNEL: =================================
# KERNEL: TC8 : BRANCH/JUMP
# KERNEL: =================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
```

```
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL:
# KERNEL: ================================
# KERNEL: RISC_TOP VERIFICATION COMPLETE
# KERNEL: ================================
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# KERNEL: ASSERT PASS : RESET PC
# KERNEL: ASSERT PASS : PC UPDATE
# KERNEL: ASSERT PASS : STALL
# KERNEL: ASSERT PASS : WRITEBACK
# RUNTIME: Info: RUNTIME_0068 testbench.sv (416): $finish called.
# KERNEL: Time: 1115 ns,  Iteration: 0,  Instance: /tb_risc_top,  Process: @INITIAL#235_1@.
# KERNEL: stopped at time: 1115 ns
# VSIM: Simulation has finished. There are no more test vectors to simulate.
acdb save
acdb report -db fcover.acdb -txt -o cov.txt -verbose
# ACDB: Coverage report "cov.txt" was generated successfully.
```

```
exec cat cov.txt
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       REPORT INFO       ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# SUMMARY
# ===============================================
# |    Property    |          Value          |
# ===============================================
# | User           | runner                  |
# | Host           | 2a77f690e7e7            |
# | Tool           | Riviera-PRO 2025.04     |
# | Report file    | /home/runner/cov.txt    |
# | Report date    | 2026-06-19 06:23        |
# | Report arguments | -verbose              |
# | Input file     | /home/runner/fcover.acdb |
# | Input file date | 2026-06-19 06:23       |
# | Number of tests | 1                      |
# ===============================================
#
#
# TEST DETAILS
# ==============================================
# | Property |            Value            |
# ==============================================
# | Test     | fcover.acdb:fcover           |
# | Status   | Ok                           |
# | Args     | asim +access+r               |
# | Simtime  | 1115000 ps                   |
# | Cputime  | 0.143 s                      |
# | Seed     | 1                            |
```

```
# | Date     | 2026-06-19 06:23                |
# | User     | runner                      |
# | Host     | 2a77f690e7e7                   |
# | Host os  | Linux64                       |
# | Tool     | Riviera-PRO 2025.04 (simulator) |
# ================================================
#
#
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++     DESIGN HIERARCHY    ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
#
#
# CUMULATIVE SUMMARY
# ==============================================
# |    Coverage Type    | Weight | Hits/Total |
# ==============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# ==============================================
# CUMULATIVE INSTANCE-BASED COVERAGE: 100.000%
# COVERED INSTANCES: 1 / 1
# FILES: 1
#
#
# INSTANCE - /tb_risc_top : work.tb_risc_top
#
#
#    SUMMARY
#
==========================================================================
==
```

```
#     |    Coverage Type     | Weight | Local Hits/Total | Recursive Hits/Total |
#
======================================================================
==
#     | Covergroup Coverage |      1 |         100.000% |             100.000% |
#     |---------------------|--------|------------------|----------------------|
#     | Types               |        |            1 / 1 |                1 / 1 |
#
======================================================================
==
#     WEIGHTED AVERAGE LOCAL: 100.000%
#     WEIGHTED AVERAGE RECURSIVE: 100.000%
#
#
#     COVERGROUP COVERAGE
#
======================================================================
=====
#     |                Covergroup                |   Hits   |  Goal /  | Status  |
#     |                                          |          | At Least |         |
#
======================================================================
=====
#     | TYPE /tb_risc_top/risc_top_cg            | 100.000% | 100.000% | Covered |
#
======================================================================
=====
#     | INSTANCE RISC_TOP_COVERAGE               | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_pcsrc   | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | bin not_taken                            |      109 |        1 | Covered |
#     | bin taken                                |        1 |        1 | Covered |
#     |------------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_flush   | 100.000% | 100.000% | Covered |
#     |------------------------------------------|----------|----------|---------|
```

```
#     | bin no_flush                          |      109 |        1 | Covered |
#     | bin flush                             |        1 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_stall    | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin normal                            |      109 |        1 | Covered |
#     | bin stall                             |        1 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_forwardA | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_forward                        |      105 |        1 | Covered |
#     | bin ex_forward                        |        2 |        1 | Covered |
#     | bin mem_forward                       |        3 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_forwardB | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_forward                        |      106 |        1 | Covered |
#     | bin ex_forward                        |        3 |        1 | Covered |
#     | bin mem_forward                       |        1 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_regwrite | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin write_off                         |       11 |        1 | Covered |
#     | bin write_on                          |       99 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_memread  | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_read                           |      108 |        1 | Covered |
#     | bin read                              |        2 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_memwrite | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
```

```
#     | bin no_write                          |       109 |         1 | Covered |
#     | bin write                             |         1 |         1 | Covered |
#
==============================================================================
=====
# +++++++++++++++++++++++++++++++++++++++++++++
# ++++++++++       DESIGN UNITS      ++++++++++
# +++++++++++++++++++++++++++++++++++++++++++++
# CUMULATIVE SUMMARY
# =============================================
# |    Coverage Type    | Weight | Hits/Total |
# =============================================
# | Covergroup Coverage |      1 |   100.000% |
# |---------------------|--------|------------|
# | Types               |        |      1 / 1 |
# =============================================
# CUMULATIVE DESIGN-BASED COVERAGE: 100.000%
# COVERED DESIGN UNITS: 1 / 1
# FILES: 1
#
#
# MODULE - work.tb_risc_top
#
#
#     SUMMARY
#     =============================================
#     |    Coverage Type    | Weight | Hits/Total |
#     =============================================
#     | Covergroup Coverage |      1 |   100.000% |
#     |---------------------|--------|------------|
#     | Types               |        |      1 / 1 |
#     =============================================
#     WEIGHTED AVERAGE: 100.000%
```

```
#
#
#    COVERGROUP COVERAGE
#
==========================================================================
=====
#    |              Covergroup              |   Hits   |  Goal /  | Status  |
#    |                                      |          | At Least |         |
#
==========================================================================
=====
#    | TYPE /tb_risc_top/risc_top_cg            | 100.000% | 100.000% | Covered |
#
==========================================================================
=====
#    | INSTANCE RISC_TOP_COVERAGE               | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT RISC_TOP_COVERAGE::cp_pcsrc    | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin not_taken                            |      109 |        1 | Covered |
#    | bin taken                                |        1 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT RISC_TOP_COVERAGE::cp_flush    | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin no_flush                             |      109 |        1 | Covered |
#    | bin flush                                |        1 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT RISC_TOP_COVERAGE::cp_stall    | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | bin normal                               |      109 |        1 | Covered |
#    | bin stall                                |        1 |        1 | Covered |
#    |------------------------------------------|----------|----------|---------|
#    | COVERPOINT RISC_TOP_COVERAGE::cp_forwardA | 100.000% | 100.000% | Covered |
#    |------------------------------------------|----------|----------|---------|
```

```
#     | bin no_forward                          |      105 |        1 | Covered |
#     | bin ex_forward                          |        2 |        1 | Covered |
#     | bin mem_forward                          |        3 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_forwardB | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_forward                          |      106 |        1 | Covered |
#     | bin ex_forward                          |        3 |        1 | Covered |
#     | bin mem_forward                          |        1 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_regwrite | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin write_off                          |       11 |        1 | Covered |
#     | bin write_on                           |       99 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_memread  | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_read                            |      108 |        1 | Covered |
#     | bin read                               |        2 |        1 | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | COVERPOINT RISC_TOP_COVERAGE::cp_memwrite | 100.000% | 100.000% | Covered |
#     |----------------------------------------|----------|----------|---------|
#     | bin no_write                           |      109 |        1 | Covered |
#     | bin write                              |        1 |        1 | Covered |
#
===========================================================================
=====
#
exit
# KERNEL:
# KERNEL: Final Coverage = 100.00%
# VSIM: Simulation has finished.
```